

Extended Methods and Results

When Valid Evidence Goes Unused in Oncology Agents

Expanded appendices and supplementary clarifications

The original bibliography and expanded Appendices A to F occupy PDF pages 2 to 32, with their rendered content and printed numbers unchanged. The submitted manuscript is also unchanged.

The addendum begins on PDF page 33. It summarizes paired outcomes from existing common-prompt outputs and explains the study scope, prompt changes, selected explanation examples, and deterministic preservation. We ran no new model generations. The examples do not provide a complete qualitative census or establish an internal causal mechanism.

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Numbering and materials

For the archival body, PDF page equals original printed page minus eight. The addendum uses printed pages S1 onward. References to main sections and figures refer to the submitted manuscript; compact and archival table numbering can differ.

Finding the paper’s references. The paper uses condensed appendices; this PDF retains the expanded numbering. For example, paper Table 6 corresponds to Table 23 here (PDF page 23), and paper Table 8 to Table 31 (PDF page 28). Section S6 (PDF page 37 onward) and the repository README map the paper’s references to the relevant sections, tables, and pages.

File availability. Paths beginning with `versions/` or `analysis_inputs/`, and the directory map in original Table 13, refer to the research workspace. Those directories and the separate companion report are not included in this repository. The relevant methods and disclosed prompts can be read in this PDF; the original executable files are not supplied.

The repository contains the original supplement in `archive/`, the addendum PDF and LaTeX sources, 90 pairs of outcome labels, and scripts to recompute those counts and build the PDFs. The labels do not support independent response scoring or reproduction of the reported bootstrap intervals. Licensed candidate text, full transcripts, and private reader forms are not distributed.

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Appendix	Purpose	Main-text connection
A	Background from the original agent studies and scope errors	Introduction
B	Shared data, eligibility, retrieval, and execution foundations	Section 2
C	Fixed exposure, live retry, and earlier repair studies	Sections 3, 4.3
D	Detailed specification of the fresh/repair/reset comparison	Section 4
E	Inventory study and serialization on the same retrieved records	Section 5.1
F	Source-interpretation examples and validation scope	Sections 5.2, 6.4

A Historical Audits and Original Agent Studies

This appendix supports the Introduction’s distinction between record-level eligibility errors and evidence availability, including the provenance of the 68/352 typed-hybrid scope violations.

A.1 Prototype and Current Study Context

The historical prototype used a different corpus, traversal-based labels, and a tool that returned a final action and assertion identifiers without source therapy or response prose. Its action scores therefore are not directly comparable with the evidence-only results below. Detailed prototype counts and policy audits remain in the companion technical report’s historical-audit appendix. The Introduction’s 68/352 scope violations instead come from the typed-output study in Section [A.4](#).

A.2 Study Design and Scoring

The text-output experiment and subsequent typed-output follow-up each contain 576 trajectories: four arms on the same 48 cases, with three repetitions per arm. The cases comprise 24 historical and 24 newly sampled cases, spanning 43 variant clusters. There are 27 encoded-positive cases and 21 encoded-negative cases, yielding 81 positive and 63 negative trajectories per arm. Sampling and within-study execution order use seed 20260924. The one-search arm was added before agent execution; typed finalization was declared after completion of the text experiment. The two interfaces were executed sequentially, so their comparison describes outcomes under successive protocols rather than a randomized interface intervention.

All arms receive the same case information, source-curated variant categories, disease ancestors, evidence fields, and eligibility instructions. Hybrid one-search permits one executed search; iterative hybrid, post-filter, and scope-first permit up to five, returning at most ten records per call. Post-filtering checks the top-ranked candidates; scope-first selects from the eligible set before the return limit. Each trajectory allows eight model turns and 1,800 output tokens per response, using `claude-sonnet-5`, medium effort, and provider-default temperature. Corpus construction and ranking details appear in Appendix [B](#).

Strict correctness requires a valid schema, the correct answer/abstain action for the encoded case, and every submitted claim to pass retrieved-identifier, eligibility, and copied-therapy/direction checks. A positive answer needs at least one claim; any invalid claim invalidates the answer. The historical metric *supported positive-case coverage* is therefore valid yield on the 81 positive trajectories, not eligible exposure or inventory recall. A correct abstention contributes to strict correctness on the 63 negative trajectories but not to positive-case valid yield. Free-text reasoning is outside these structured-claim checks.

Reported 95% intervals use 2,000 percentile bootstrap draws with seed 20260924. Each draw resamples variant clusters and retains every member case and repetition; paired contrasts preserve the shared case–repetition pairs. Overall metrics use 43 clusters and positive-case yield uses 26. The bootstrap pools the retained trajectory observations within each draw; balanced repetitions also make the point estimates equal to averages of case means. Shared source publications remain a further dependence. These original-study intervals differ from the 10,000-draw analyses used in the later repair studies.

Output formats and scoring. *Raw text* uses the original strict JSON parser. *Normalized text* is a post hoc sensitivity analysis on the same outputs: remove exactly one complete outer Markdown code fence, then apply the same parser and claim checks without changing fields or repairing truncated JSON. Prose outside the JSON object remains invalid. *Typed output* comes from the separately executed follow-up, which submits the same action, claims, and reasoning fields through `submit_inventory` with required tool use. Normalization changes scoring of saved text; typed finalization changes the output interface in a later study.

A.3 Text-Output Results and Parser Sensitivity

All 576 text trajectories completed without execution failures or retries. Under the prespecified strict JSON parser, strict correctness is 71.5% for both one-search and iterative hybrid, 76.4% for post-filter, and 66.7% for scope-first (Table 2). Positive-case valid yield is 54.3%, 84.0%, 82.7%, and 63.0%, respectively. Thus, scope-first has lower strict correctness and valid yield than iterative hybrid in this original text-output experiment.

Fence normalization makes 54 additional outputs parseable, leaving 84 invalid schemas. Table 2 distinguishes raw from normalized scoring; Table 3 reports the normalized historical/new-case split.

Table 2: Text-output study: 48 cases $\times$ three repetitions per arm. The raw-scoring columns use the prespecified parser; the normalized-scoring group removes one complete outer code fence without field or claim repair. Strict correctness requires the correct action and support for every claim. Positive-case valid yield uses 81 trajectories per arm; Invalid counts answers containing at least one invalid claim. Intervals preserve repetitions within variant clusters.

Arm	Raw text scoring		Normalized text scoring		
	JSON (%)	Strict (%)	Strict (%) 95% CI	Valid yield (%)	Invalid answers
Hybrid, one search	87.5	71.5	79.9 [68.7,89.5]	69.1	0
Hybrid, iterative	72.9	71.5	76.4 [67.3,84.9]	92.6	3
Hybrid, post-filter	77.1	76.4	84.0 [75.5,91.5]	96.3	0
Hybrid, scope-first	66.7	66.7	82.6 [74.1,90.6]	91.4	0

Table 3: Text-output results by case source after outer-fence normalization. Each source comprises 24 cases and three repetitions (72 trajectories). Valid yield uses 54 historical positive trajectories or 27 new positive trajectories. Scope errors count claims failing encoded eligibility.

Arm	Source	Runs	Normalized strict (%)	Normalized valid yield (%)	Normalized scope errors
Hybrid, one search	legacy	72	79.2	75.9	0
Hybrid, one search	new	72	80.6	55.6	0
Hybrid, iterative	legacy	72	77.8	90.7	1
Hybrid, iterative	new	72	75.0	96.3	3
Hybrid, post-filter	legacy	72	93.1	94.4	0
Hybrid, post-filter	new	72	75.0	100.0	0
Hybrid, scope-first	legacy	72	88.9	87.0	0
Hybrid, scope-first	new	72	76.4	100.0	0

Iterative hybrid increases positive-case valid yield over one-search hybrid by 29.6 percentage points [11.1, 47.4] under strict parsing, while overall strict correctness is unchanged. After fence normalization, the yield difference is 23.5 points [5.5, 42.3]. Normalized strict correctness for scope-first minus iterative hybrid is +6.2 points [−2.8, 16.3]; the interval includes both lower and higher correctness. Formatting sensitivity changes the observed ordering without replacing the original endpoint.

A.4 Typed-Output Results and Scope Errors

All 576 typed trajectories satisfy the schema, with no execution failures, retries, or parsing repairs. Both filtered arms attain 143/144 strict-correct trajectories and 80/81 positive-case valid answers.

Iterative hybrid attains 112/144 and 65/81, respectively (Table 4). Its 352 submitted claims all have retrieved identifiers and exactly copied therapy and direction fields, yet 68 violate encoded eligibility, affecting 28/144 trajectories. This claim-level audit supplies the introduction’s 68/352 result. The v38 E0 reconstruction regraded the saved v14 typed transcripts and reproduced the recorded scores.

Table 4: Typed-output results: 48 cases and three repetitions per arm. All 144 outputs per arm satisfy the schema. Strict correctness includes valid abstention on negatives; valid yield uses 81 positive trajectories. Scope columns count invalid claims and affected submissions. All retrieved-identifier and copied-field checks pass. Intervals are 95% variant-cluster bootstrap intervals.

Arm	Strict	Strict (%) 95% interval	Valid yield (%)	Scope claims	Scope runs
Hybrid, one search	115/144	79.9 [67.4,90.2]	67.9	6/149	3/144
Hybrid, iterative	112/144	77.8 [68.6,85.6]	80.2	68/352	28/144
Post-filter, iterative	143/144	99.3 [97.8,100.0]	98.8	0/371	0/144
Scope-first, iterative	143/144	99.3 [97.8,100.0]	98.8	0/531	0/144

The typed scope-first-minus-iterative-hybrid difference in strict correctness is +21.5 percentage points [13.5, 31.2]. Iterative hybrid minus one-search hybrid increases positive-case valid yield by 12.3 points [−2.5, 28.6]. The two filtered arms have equal aggregate correctness and equal case-mean correctness, but differ on two repetitions of case N0116. Their paired correctness difference and bootstrap interval are 0.0 [0.0, 0.0] because the within-case differences cancel when clusters retain all repetitions. This observed equality is not a population-equivalence test. Scope-first uses 0.58 fewer searches per trajectory [0.32, 0.87] than post-filter, while using 6.7% more input tokens and 7.6% more output tokens.

A.5 Error Units and Overlapping Categories

Table 5 distinguishes invalid schemas, explicit positive-case abstentions, negative-case affirmative answers, and invalid structured claims. It retains the original strict text scoring alongside the fence-normalized sensitivity and the typed follow-up. Schema failures receive no inferred biological action. Claims are counted only when parsing succeeds; affirmative-action and claim failures can overlap. All parsed claims in these studies pass provenance and copied-field checks, so their observed claim errors concern encoded eligibility. The normalized text-hybrid row has four scope-invalid claims in three answers, compared with three claims in two answers under strict parsing; this change reflects the additional parsed output.

These are audits of final agent submissions. The original runner scores the completed output after generation; it does not return rejection feedback and execute a repair-and-release cycle. Consequently, the invalid answers here are not counts of errors released after a blocking checker. Attempted and released errors in the later repair protocols are reported separately in Appendix C. Throughout this module, validity concerns the encoded conditions and copied fields; source-level biological interpretation has a separate evaluation contract.

A.6 Resource Use and Repeated Outcomes

Tables 6 and 7 retain the resource and repetition details for both interfaces. Searches count executed retrieval calls and exclude terminal inventory submission. Model turns count provider responses; one turn can request multiple searches. Requests made after observing a result are recorded separately, including requests refused after the search budget is exhausted. Wall time is measured under the original six-worker execution with a shared retrieval lock. Token totals are provider-reported usage across every trajectory, not a monetary estimate.

A.7 Retrieval Behavior and an Example Trace

Typed iterative hybrid provides eligible exposure on the first search in 55/81 positive trajectories and during any search in 72/81. Typed post-filter reaches eligible evidence after an empty first search in 26 positive trajectories, giving 81/81 exposure across its complete search paths. Scope-first provides eligible exposure on the first search in all 81 positives. Every trajectory executes at least one search.

Table 5: Mixed units with overlapping categories: Format counts outputs, Scope counts claims, and FP can overlap with Invalid answers; these columns must not be summed as mutually exclusive terminal outcomes. Each row covers 144 trajectories: 81 positives and 63 negatives. Format = invalid schema; FN = explicit positive-case abstention; FP = negative-case answer; Invalid = answer containing an invalid claim; Scope = claims failing eligibility. FN, FP, and Invalid count outputs; Claims counts parsed claims. Claims are counted only in schema-valid outputs. All provenance and copied-therapy/direction error counts are zero. Malformed outputs receive no inferred action. Normalized rows apply only the outer-fence sensitivity.

Scoring	Arm	Format	FN	FP	Invalid	Scope	Claims
Text, strict	One search	18	23	0	0	0	95
Text, strict	Iterative	39	0	2	2	3	220
Text, strict	Post-filter	33	1	0	0	0	192
Text, strict	Scope-first	48	0	0	0	0	241
Text, normalized	One search	6	23	0	0	0	155
Text, normalized	Iterative	31	0	2	3	4	252
Text, normalized	Post-filter	22	1	0	0	0	243
Text, normalized	Scope-first	25	0	0	0	0	376
Typed	One search	0	26	3	3	6	149
Typed	Iterative	0	4	16	28	68	352
Typed	Post-filter	0	1	0	0	0	371
Typed	Scope-first	0	1	0	0	0	531

Table 6: Original agent resources. Each row covers 144 trajectories. Tokens are totals; searches and turns are means per trajectory; wall time is the median in seconds.

Interface	Arm	Input tokens total	Output tokens total	Searches/run mean	Model turns/run mean	Wall time (s) median
Text	One search	699,385	53,459	1.00	2.01	6.1
Text	Iterative	2,901,476	134,090	3.20	3.17	12.8
Text	Post-filter	813,096	79,337	3.55	3.36	9.7
Text	Scope-first	1,134,583	92,118	3.31	3.25	10.3
Typed	One search	812,665	55,215	1.00	2.00	6.8
Typed	Iterative	3,288,834	96,602	3.25	3.41	12.1
Typed	Post-filter	1,108,850	82,544	3.71	3.65	10.6
Typed	Scope-first	1,182,598	88,843	3.13	3.23	10.2

Table 7: Strict correctness by repetition (%; 48 trajectories per repetition) and mean subsequent search requests per trajectory. Text correctness uses the declared fence-normalized sensitivity; typed correctness uses the original schema. Subsequent requests are issued after at least one search result has been observed and need not be executed.

Scoring	Arm	Repetition 1 strict (%)	Repetition 2 strict (%)	Repetition 3 strict (%)	Subsequent requests/run mean
Text, normalized	One search	77.1	79.2	83.3	0.01
Text, normalized	Iterative	72.9	75.0	81.2	1.52
Text, normalized	Post-filter	81.2	83.3	87.5	2.12
Text, normalized	Scope-first	81.2	87.5	79.2	1.89
Typed	One search	81.2	79.2	79.2	0.00
Typed	Iterative	68.8	79.2	85.4	1.94
Typed	Post-filter	97.9	100.0	100.0	2.54
Typed	Scope-first	100.0	97.9	100.0	1.93

452 These logged observations distinguish evidence becoming available from its inclusion in the final
453 submission.

454 The post hoc trace audit selects the lexicographically first successful typed iterative-hybrid trajectory
455 with no eligible record in the first search and an eligible record in a later search: NCOA4–RET in lung
456 adenocarcinoma, case B039, repetition two. Exact-fusion and drug-qualified searches each return ten
457 records with no eligible match. A third query uses the source RET-fusion category and returns five
458 eligible records among ten. This selection illustrates observation-dependent query reformulation; it is
459 a single selected trace rather than an additional performance sample.

460 These original studies distinguish eligibility errors in submitted claims from eligible exposure during
461 search. Recovery after a citation is rejected is evaluated separately in the repair protocols (Appendix C)
462 and the common-prompt fresh/repair/reset comparison (Appendix D).

463 **B Corpus, Retrieval, and Execution Details**

464 This appendix supports Section 2 by specifying the shared corpus, encoded eligibility checks,
465 exposure measures, and execution conditions.

466 **B.1 Corpus and Case Selection**

467 The shared corpus contains 15,167 CKB evidence records from release 260403, with disease ancestry
468 from Disease Ontology release 2026-08-31. It covers EGFR, ALK, ROS1, BRAF, KRAS, MET,
469 RET, ERBB2, NTRK1, and TP53 and retains all 6,078 records from the historical dense baseline.
470 An additional 6,839 panel-related corpus rows were excluded because their disease identifiers were
471 absent from the frozen ontology, representing 31.1% of the combined included and excluded rows.
472 This is a corpus-coverage exclusion: these rows were outside the evaluated corpus, rather than labeled
473 case-ineligible records.

474 Variant resolution uses exact source identifiers or names, with lossless conversion of three-letter
475 to one-letter amino-acid substitutions. Ambiguous source labels and unresolved identities remain
476 unresolved. The procedure performs neither general VCF normalization nor transcript disambiguation.
477 Records preserve their source molecular profiles and disease identifiers, with DO aliases resolved
478 against the frozen release. Category membership requires a source-curated path linking the variant to
479 that category.

480 The new retrieval sample comprises 240 specific source variants, each paired with lung adenocar-
481 cinoma (LUAD) and lung squamous cell carcinoma (LUSC): 480 synthetic single-variant, somatic
482 cases, including 396 encoded positives and 84 encoded negatives. Sampling used seed 20260924
483 before retrieval outcomes were inspected. Within each gene, available substitution, indel, fusion, and
484 other descriptor strata were sampled in round-robin order, up to 24 variants per gene. Source category
485 and non-specific variant entries, variants used in the historical sample, and duplicates already selected
486 for another gene were excluded from this sampling pool. Selection did not condition on evidence
487 counts or retrieval outcomes.

488 The historical queries yield 116 distinct cases after deduplication. Of these, 17 have unresolved variant
489 identities and are excluded from labeled accuracy denominators; they are not encoded negatives.
490 The remaining 99 comprise 88 encoded positives and 11 encoded negatives. This case-resolution
491 exclusion is separate from the corpus-row exclusion above.

492 The original agent subset selects 24 historical and 24 newly sampled cases, balancing available
493 action and descriptor strata before model execution. These same 48 cases—27 positives and 21
494 negatives in 43 variant clusters—are reused by the text-output, typed-output, and inventory studies
495 (Appendices A.3 and E). The first two each use three repetitions in four arms (576 trajectories per
496 study); inventory uses three repetitions in three arms (432 trajectories). Repetitions and arms add
497 trajectories, not independent cases.

498 The fixed-exposure study (Appendix C) reuses 45 earlier positive repair cases: 32 co-alteration and
499 13 sibling-disease cases. The common-prompt study (Appendix D) reuses all 45 with the same
500 ordered candidate records. Fixed exposure additionally includes 12 encoded-negative cases and 12
501 genuine-pass controls that reuse positive cases. Live retry (Appendix C) uses a separately selected
502 set of 24 co-alteration positives, 12 sibling-disease positives, and 12 negatives. Its positive variant

families were absent from the original agent subset and earlier repair proposals; 11 negative queries were reused and one was new. Some positive candidate publications overlap earlier retrieved payloads, so this is case-family separation rather than publication-disjoint evaluation.

B.2 Eligibility Rules and Submission Checks

Encoded eligibility policy. All eligibility calculations implement Equation 1. The indexed implementation computes the ancestor closure of observed variants and the query disease, intersects inverted indices, then checks every required molecular-profile component and the origin restriction. Co-alteration profiles are conjunctive: every encoded component must be covered. Disease matching admits the query disease and its ancestors; sibling and narrower diseases outside that path fail. An unreported molecular condition remains unconfirmed. Table 8 gives the origin policy, including source-unspecified values. Biological interpretation additionally checks origin and other qualifiers in the publication.

Implementation conformance. A separate reference scanner computes membership directly from raw source category paths and DO ancestry, independently of retrieval. An in-memory SQLite implementation evaluates molecular containment, disease ancestry, and origin with relational queries. Indexed and relational implementations return identical sets on all 579 resolved cases. Agreement tests implementation conformance to the shared encoded conditions; publication-level support is assessed separately (Appendix F). Regression cases cover same-codon sibling alleles, sibling and narrower diseases, co-alterations, origin restrictions, unknown identities, citation provenance, copied fields, and malformed affirmative outputs.

Submission checking. The structured checker tests each submitted identifier against the records supplied to that trajectory, verifies case eligibility, and requires exact therapy and direction fields. A valid answer has a valid schema and at least one claim, with every claim passing these checks. Strict correctness additionally requires the appropriate answer or abstain action for the encoded case label; it therefore includes valid abstentions on negatives. Claim-level violations, invalid affirmative submissions, positive-case abstentions, and invalid released claims have separate denominators. Generated explanations are outside these structured checks. Repair feedback and release policies are specified in Appendices C and D.

Table 8: Origin restrictions in the encoded eligibility policy. Stored evidence values are lowercased and stripped of surrounding whitespace; missing values become empty strings. An unconstrained value passes this check while retaining the source value for later interpretation.

Normalized evidence-origin value	Admission rule
Empty, any, both, mixed, unknown, not specified, n/a	Pass the origin check (unconstrained).
somatic	Require case origin somatic.
germline	Require case origin germline.
Any other value	Fail the origin check.

B.3 Ranking, Candidate Budgets, and Filter Placement

All retrieval methods score the same complete corpus and use evidence-identifier order to break ties. Ranking text serializes the evidence identifier, molecular profile, therapy, indication, response direction, evidence type and level, origin, and source prose in that order. Tool payloads expose these evidence fields and publication identifiers, without eligibility labels, reference sets, or an expected final action.

Dense retrieval uses normalized 768-dimensional S-PubMedBERT-MS-MARCO embeddings (Deka et al., 2022), pinned to revision 96786c7024f95c5aac7f2b9a18086c7b97b23036, with a 512-token encoding limit. BM25 uses the full serialized text, $k_1 = 1.2$, and $b = 0.75$ (Robertson and Zaragoza, 2009). Hybrid retrieval combines complete dense and lexical rankings through reciprocal-rank fusion with constant 60 (Cormack et al., 2009). The exploratory expanded query appends the source variant-ancestor labels and disease-ancestor names once in deterministic order. Ranker parameters remain fixed across comparisons.

For a ranking $R(q)$, post-filtering first takes k ranked candidates and returns $R_k(q) \cap G(q)$: k is the candidate count before filtering. Scope-first scans the same ranking for members of $G(q)$ and returns at most k eligible records: k is a return limit. Both use whole-corpus scores, while the selection step differs. For a positive case and $k \geq 1$, scope-first coverage is 100% by construction and eligible-ID recall is $\min(k, |G(q)|)/|G(q)|$. Equal numerical k therefore specifies different limits; candidate depth, returned payload size, and model-token usage remain separate resource measures.

B.4 Retrieval Outcomes and Uncertainty

Retrieval coverage is the fraction of positive cases with at least one eligible returned record. Eligible-ID recall is the fraction of $G(q)$ returned, before agent finalization. Both average over 396 positive cases in 199 variant clusters. The invalid fraction averages each case’s proportion of ineligible returned records over all 480 cases, assigning zero to empty returns; invalid-pair counts instead sum case–record pairs. Negative-hit rates use the 84 encoded negatives.

At $k = 10$, unexpanded hybrid retrieval covers 41.2% of positives; filtering its top ten removes ineligible records while preserving coverage and recall (Table 9). Expanded hybrid post-filtering covers 81.3% with 20.3% eligible-ID recall. Scope-first returns up to ten eligible records, with 66.1% recall implied by the eligible-set sizes. Table 10 gives the unexpanded hybrid results by descriptor class. On the separate 88 resolved historical positives, dense, BM25, and hybrid coverage at ten candidates is 36.4%, 34.1%, and 51.1%, respectively.

Table 9: Retrieval at $k = 10$ on 480 new cases (396 positives). Here k limits candidates before post-filtering, but returned eligible records for scope-first; equal k is not a matched payload or token budget. Coverage and eligible-ID recall use the 396 positive cases. Invalid is the mean percentage of case-ineligible records across all cases, with empty outputs assigned zero. Brackets denote variant-cluster 95% intervals (2,000 draws for original rows; 10,000 for expanded post-filter); the last column counts case-ineligible query–record pairs. Scope-first coverage of 100% and filtered encoded violations of 0% are guaranteed by Equation 1. *Exploratory ontology query expansion without filtering.

Method	Coverage [95% CI]	Recall	Invalid (%)	Invalid pairs
Dense	34.6 [28.7,40.5]	4.2	95.4	4580
BM25	18.9 [14.2,23.5]	2.1	97.5	4682
Hybrid RRF	41.2 [35.4,46.7]	4.5	94.1	4519
Hybrid + post-filter	41.2 [35.4,46.7]	4.5	0.0	0
Expanded hybrid + post-filter	81.3 [76.3,86.1]	20.3	0.0	0
Hybrid + scope-first	100.0 (by construction)	66.1	0.0	0
Dense + ontology terms*	44.9 [38.7,51.6]	5.4	92.8	4456
BM25 + ontology terms*	64.6 [58.0,70.6]	8.9	86.3	4141
Hybrid + ontology terms*	81.3 [76.1,86.1]	20.3	78.1	3748

Table 10: Descriptor-stratified hybrid retrieval at $k = 10$ on the new sample, with the original unexpanded query. Coverage uses positive cases; invalid fraction uses all cases in a class.

Class	Cases	Positives	Coverage (%)	Invalid (%)
substitution	196	136	36.8	94.9
indel	138	128	30.5	95.2
fusion	142	132	56.1	91.8
other	4	0	–	100.0

The post hoc expanded-ranking replay evaluates $k \in \{5, 10, 20, 40, 80, 160\}$. Applying the predicate after each cutoff preserves expanded-ranking coverage and recall in all 2,880 case–cutoff combinations while removing ineligible returns. At depth 160, expanded post-filtering reaches every sampled positive and returns 63.3% of the eligible IDs on average (Table 11). The figure compares ordinary and expanded post-filtering with scope-first selection under these distinct candidate and return limits.

Original retrieval and query-expansion intervals use 2,000 variant-cluster bootstrap draws with seed 20260924. The expanded post-filter replay and Figure 5 use 10,000 draws with seed 20260927. Resampling keeps disease-paired queries for a variant together; all-case measures use 240 variant

Table 11: Post-hoc expanded-hybrid post-filter sweep. Coverage/recall (%) use 396 positives; returned counts average all 480 cases. Zero encoded violations follow by construction from filtering every returned ID with the scope predicate. Candidate depth is the pre-filter cutoff and differs from returned payload size.

Candidate k	Coverage	Recall	Returned	Invalid pairs
5	66.2	14.0	1.39	0
10	81.3	20.3	2.19	0
20	87.4	26.3	3.17	0
40	93.7	35.4	4.83	0
80	97.7	51.0	7.82	0
160	100.0	63.3	11.38	0

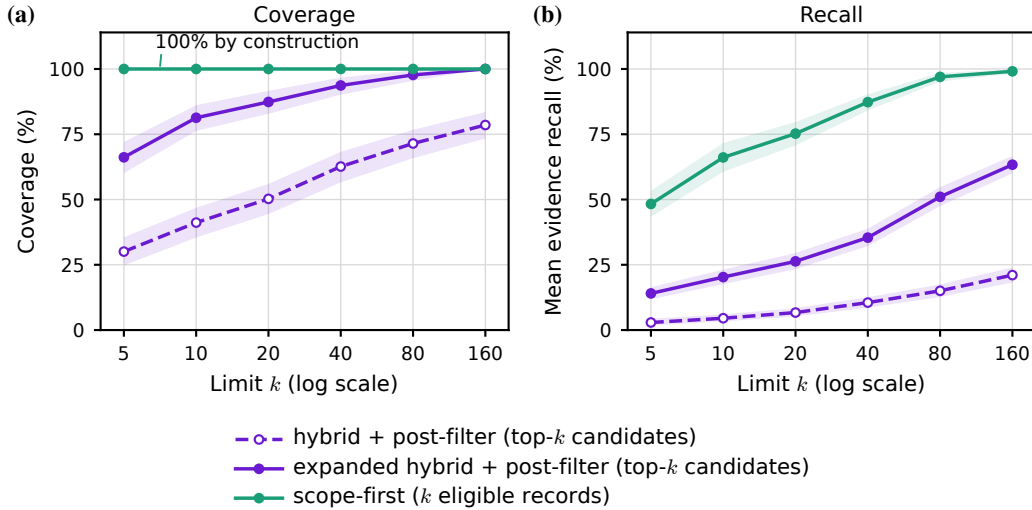


Figure 5: Eligible evidence exposure over 396 encoded-positive cases. (a) Positive-query coverage and (b) mean eligible-ID recall for ordinary hybrid post-filtering, expanded hybrid post-filtering, and scope-first admission. Bands are 95% variant-cluster bootstrap intervals from the saved replay (199 positive variant clusters, 10,000 draws). Scope-first coverage is 100% by construction and has no sampling band. The limit k is the pre-filter candidate depth for post-filtering and the returned-record limit for scope-first; the logarithmic axis doubles k at each tick. Returned payload sizes and token costs differ between policies.

clusters. The differing bootstrap draws account for the small interval differences between the expanded filtered and unfiltered rows despite equal point estimates. Complete per-case sweeps and paired contrasts are retained in the analysis artifact.

With every admitted record returned, exact variant and disease matching covers 5.3% of positives (Table 12); allowing only disease ancestry or only variant ancestry gives 13.6% and 47.7%, respectively. Removing disease restrictions returns records for 38.1% of negatives. Replacing molecular containment with gene overlap returns records for every negative and produces a 93.6% mean invalid fraction. These comparisons describe omissions and additions relative to the fixed reference predicate.

B.5 Execution Settings, Runtime, and Artifact Access

Across the 579 resolved cases, median warm indexed admission takes 0.135 ms, and the SQLite predicate takes 6.223 ms. The indexed call follows an initial admission call; the timed SQLite call receives precomputed ancestor sets and includes loading its query tables. Query encoding plus dense/BM25/RRF scoring takes a median 37.3 ms. These component timings come from a shared CPU host. The admission timings cover the eligibility operation; end-to-end retrieval also includes corpus-wide ranking, and agent execution adds model calls.

Table 12: Predicate ablations on the same new cases and corpus, returning every admitted record without a ranking cutoff. Coverage and eligible-ID recall use the 396 positives; invalid fraction uses all 480 cases, with empty returns assigned zero. Negative-hit rate is the percentage of 84 negative queries returning any record. All values are percentages. Exact matching omits case-eligible records; permissive matching admits case-ineligible records. The “Both source hierarchies” row enumerates the Equation 1 reference set, so its 100% coverage and recall are guaranteed by construction.

Predicate	Coverage	Recall	Invalid	Negative hit
Exact variant + exact disease	5.3	1.0	0.0	0.0
Exact variant + disease ancestry	13.6	2.9	0.0	0.0
Variant ancestry + exact disease	47.7	4.5	0.0	0.0
Both source hierarchies	100.0	100.0	0.0	0.0
Remove disease predicate	100.0	100.0	54.5	38.1
Gene overlap + disease ancestry	100.0	99.6	93.6	100.0

All agent and repair studies use the requested and returned alias `claude-sonnet-5`, medium reasoning effort, and provider-default temperature. No immutable model snapshot was available. Original text-output and typed-output arms receive the same case, source variant categories, disease ancestors, and evidence fields, with one or at most five searches, ten records per search, eight model responses, and 1,800 output tokens per response. Multiple searches within a response each consume one search-budget unit; over-budget requests return an error without records. The original runners permit up to three recorded transient retries after the initial request. The two output-interface studies ran sequentially, so their comparison includes execution-time differences (Appendix A.3).

The later studies freeze their own submission budgets, output caps, interleaving, and failure-handling policies. Fixed exposure and common-prompt context comparisons use a 1,800-token response cap; inventory and live retry use 4,096. Their complete exceptions and resources appear in Appendices C, D, and E. Invalid outputs and execution failures remain in the planned outcome denominators. Throughout the paper, *prespecified* means locally frozen with a protocol and hash manifest; retrospective replays retain their post hoc designation.

Recorded request-start windows are 24 September 2026, 13:11–13:28 UTC for text output and 13:55–14:03 UTC for typed output, and 26 September 2026, 20:38–20:54 UTC for inventory. The timing supplement at `versions/v39-agenticls-scope-repair-paper/analysis_inputs/study_timing_supplement.json` preserves full timestamps and returned model aliases.

Table 13 maps experiment names to repository-relative directories. These are local artifact locations; a public deposition URL is not recorded. The original-agent and inventory package is retained at `versions/v21-reviewer-controls/artifact/granite_reviewer_controls_artifact.zip`. Its boundary excludes licensed source text and full transcripts; later-study directories are listed separately.

Table 13: Local artifact directory map. Paths start at the repository root; filenames in the accompanying text are relative to the corresponding experiment directory. Internal version names identify artifacts, not independent samples.

Experiment or shared material	Directory
Shared corpus and original agents	<code>versions/v14-full-paper-revision/</code>
Explicit inventory	<code>versions/v21-reviewer-controls/</code>
Expanded retrieval replay; live retry	<code>versions/v38-agenticls-revision-experiments/ (e0/; e1/)</code>
Fixed-exposure repair	<code>versions/v40-fixed-exposure-repair/</code>
Common-prompt fresh/repair/reset	<code>versions/v41-repair-history-control/</code>
Saved-pool serialization	<code>versions/v42-saved-pool-serialization-replay/</code>

The shared corpus’s `README.md` gives reconstruction commands; `src/common.py` specifies eligibility, and `src/agents.py` specifies parsing, provenance, and copied-field scoring. Frozen prompt/tool definitions and message assembly are available in the original studies’ `runs/agent_execution_plan.json` and `runs/typed_agent_execution_plan.json`, inventory’s `runs/execution_`

plan.json, live retry’s e1/runs/execution_plan.json, fixed exposure’s src/study.py, and the common-prompt study’s design/public_prompt_contract.json. Repair protocols reside in design/protocol.md; inventory uses design/plan.md, and live retry additionally records its final selection in e1/final_feasibility_amendment.md. These paths identify the executed specifications; the study appendices explain their differences.

Saved case identifiers and per-run scores retain case, arm, and repetition membership for reconstructing aggregate results. The original studies provide runs/agent_analysis_per_run.json and runs/typed_agent_analysis_per_run.json; inventory provides analysis/fresh_per_run.json; live retry provides e1/analysis/per_run.json; fixed exposure provides runs/scores/ and runs/paired_cases.json; and the common-prompt study provides runs/per_run.json. The saved-pool replay’s src/replay.py, analysis/per_run.json, and analysis/input_manifest.json identify its operation, scores, and inputs. Shared input manifests and the later execution plans record SHA-256 hashes, case selections, and execution order.

Saved numerical summaries support table reconstruction. Rebuilding the corpus additionally requires authorized access to CKB release 260403 and the pinned ontology, checked against the shared runs/input_manifest.json; regenerating generated outputs or repeating the saved-pool replay also requires private source-bearing records and full transcripts. Those inputs remain excluded from the distributable package. The repair experiments reuse the corpus and predicate from source-table preparation, while extraction and source-judgment studies form separate analyses.

B.6 Additional Related Work

Evidence use and revision. Sufficient-context analysis distinguishes missing evidence from failure to use answerable context (Joren et al., 2025). ALCE evaluates correctness and citation quality (Gao et al., 2023); Reflexion and Self-Refine use language feedback for revision (Shinn et al., 2023; Madaan et al., 2023); ETHOS applies checks to revise or suppress clinical-agent outputs (Sharma et al., 2026). Self-RAG learns retrieval and critique decisions (Asai et al., 2024), while CRAG evaluates retrieved evidence and can trigger corrective retrieval (Yan et al., 2024). These approaches motivate the recovery question here: whether eligible evidence already in context is submitted after rejection.

Qualified matching and retrieval. Cat-VRS represents categorical matching through constraints and containment (GA4GH, 2026); VICC, MOAlmanac, and CKB encode relationships used in molecular evidence interpretation (Wagner et al., 2020; Reardon et al., 2021; Patterson et al., 2019). ACORN and Filtered-DiskANN study efficient predicate-aware approximate nearest-neighbor search (Patel et al., 2024; Gollapudi et al., 2023). Here, full-corpus ranking and explicit admission define case eligibility and control evidence exposure for the agent studies.

Record preservation and workflow scope. GraphRAG uses graph structure for retrieval and summarization (Edge et al., 2024); MAGG governs candidate-triple admission with domain-expert agents and provenance (Bykampadi et al., 2026). Our fixed store supplies records whose case eligibility is computed by indexed and relational implementations of the same predicate. MedAgent-Bench evaluates physician-written tasks in an interactive FHIR environment (Jiang et al., 2025); the present inventory task instead isolates evidence eligibility, recovery, and preservation for subsequent biological interpretation.

C Citation Repair: Fixed Exposure, Live Retry, and Earlier Studies

This appendix supports the fixed-exposure finding in Section 3 and the live-retry feedback comparison in Section 4. Table 14 distinguishes the four studies. Valid recovery requires a nonempty inventory whose entire claim set passes schema, provenance, encoded eligibility, and copied-field checks. Each study uses two repetitions per condition; prespecified analyses and decision rules were frozen locally before generation. Common model settings and reconstruction information are in Appendix B.5. Study-specific budgets, matching, and inference are detailed below. The common-prompt fresh/repair/reset comparison is specified separately in Appendix D.

Table 14: Guide to the repair studies. Checklist and identifier-only feedback are distinct comparators. Each study uses two repetitions per condition; its cases, response budget, and analysis remain separate.

Study	Comparator	Defining condition
Fixed exposure (C.1)	Checklist vs. diagnostic feedback	Same candidates; two eligible alternatives guaranteed; no search
Live retry (C.2)	Checklist vs. diagnostic feedback	Additional searches/resubmissions on a separate sample
Pilot (C.3)	Identifier-only vs. diagnostic feedback	Natural and constructed errors; feedback length uncontrolled
Crossed study (C.3)	Identifier-only/diagnostic feedback × unfiltered/scope-first retry	Earlier crossing of feedback and retry retrieval; feedback length uncontrolled

C.1 Fixed Exposure: One Revision with Eligible Alternatives

Recovery, abstention, and retention controls. Positive cases test recovery through eligible alternatives after rejection. Encoded-negative controls test behavior when eligible evidence is unavailable. Genuine-pass controls test retention of an already valid submission. The genuine-pass controls reuse positive queries, so their study-item counts are kept separate from the number of distinct cases.

Cases, candidates, and feedback. The fixed-exposure study reuses all 45 positive proposals from the earlier crossed study: 32 co-alteration and 13 sibling-disease cases. They span 39 variant clusters, including 30 in the primary co-alteration stratum and 11 in the secondary sibling-disease stratum, with overlap between strata. For each case, frozen hybrid ranking supplies the two highest-ranked eligible records, the rejected record, and the highest-ranked ineligible records needed to reach ten candidates. The mixed payload is restored to hybrid-rank order and carries no eligibility labels. The forced inclusion of two eligible records fixes exposure before the agent revises its inventory.

Both arms receive the same case, proposal, and ordered payload. The exact initial instruction and its relation to the later common prompt appear in Appendix C.4. The proposal is constructed and supplied as prior assistant/tool history. Checklist feedback names the rejected identifiers and reviews all conditions; diagnostic feedback names the same identifiers and reports the case–record checks. Neither supplies accepted replacement identifiers or a reference answer. Checklist length is selected from prefixes of a fixed checklist using only native token length. Each pair differs by at most 10% relative to diagnostic feedback; the observed maximum is 9.69%. Counts include isolated-user-message framing, without an empty-message subtraction. The model submits once, with no search and a 1,800-token output limit.

Twelve encoded-negative cases receive ten ineligible candidates including the rejected record. Twelve genuine-pass controls reuse positive queries, replace the initial proposal with an eligible claim, and receive identical pass feedback in both nominal arms. Thus, 69 study items produce 276 planned trajectories; the pass controls do not add independent biological cases. Case–repetition blocks and arm order are randomized with seed 20260927. All 276 trajectories complete without provider, schema, or truncation failures. The saved verification checks 138 identical-exposure pairs, 276 rescored responses, and 14 frozen input/code hashes. Native counting is separate from generation: 149 counting attempts include one failed preparation request.

Table 15: Length-matched fixed exposure: valid recovery with two eligible alternatives among ten visible records, no search, one revision, 1,800 output tokens, and two repetitions per arm. Differences are diagnostic minus checklist in percentage points, with 95% paired variant-cluster bootstrap intervals (30 co-alteration and 11 sibling-disease clusters). Every unrecovered positive trajectory abstains.

Family (cases)	Checklist	Diagnostic	Difference [95% interval]
Co-alteration (32)	10/64	11/64	+1.6 [0.0, 4.8]
Sibling disease (13)	3/26	0/26	-11.5 [-31.8, 0.0]

Recovery and the benefit criterion. The primary diagnostic-minus-checklist recovery difference is +1.6 percentage points (95% paired variant-cluster bootstrap interval [0.0, 4.8]): 11/64 versus 10/64.

692 The prespecified benefit criterion requires a gain of at least ten percentage points and a positive lower
693 interval bound. Repetitions are averaged within case before 10,000 paired cluster resamples (seed
694 20260927). Case-resampling sensitivity gives [0.0, 4.7] and gene-resampling sensitivity across eight
695 genes gives [0.0, 5.8] percentage points. Shared visible publications connect all 45 positive cases
696 into one component, so these intervals summarize case/variant dependence rather than independent
697 publication families.

698 Every unrecovered positive trajectory abstains. In the exploratory sibling-disease stratum, recovery is
699 3/26 with checklist feedback and 0/26 with diagnostic feedback, a difference of -11.5 percentage
700 points $[-31.8, 0.0]$. Across both positive strata, recovery is 13/90 and 11/90, respectively $(-2.2$
701 $[-7.8, 2.3])$. A descriptive case-level count finds at least one successful repetition in five of 32
702 checklist co-alteration cases and six of 32 diagnostic cases; sibling-disease recovery occurs in two
703 checklist cases and no diagnostic cases.

704 **Control outcomes and decision rules.** The negative diagnostic arm attempts two invalid affirmative
705 inventories, both repetitions of query E077, containing three invalid claims; the checker blocks all
706 three. Checklist feedback produces no such attempt. All genuine-pass controls retain the original
707 claim set (24/24 per arm). These small controls report attempted errors and retention; the common
708 release checker constrains released violations.

709 Fixed exposure fails both the primary benefit criterion and the descriptive negative-control guard.
710 The $+1.6$ -point recovery gain falls below the ten-point threshold, and its lower interval bound is
711 zero. The control guard permits zero observed increase in harmful outcomes: diagnostic feedback
712 adds two invalid-answer trajectories on negatives, while genuine-pass controls show zero increase in
713 nonrecovery.

714 **Resources and exploratory prose audit.** Across 138 trajectories per arm, checklist/diagnostic input
715 tokens total 592,120/592,894 and output tokens total 74,548/60,293. Feedback messages are length-
716 matched; the input and output totals measure complete interaction costs. A lexical screen flags 217
717 of 276 outputs; a single Codex AI reviewer examines those flagged items and 20 randomly selected
718 unflagged outputs. Among these 237 items, 48 receive ambiguous-shorthand/overstatement labels,
719 132 retain explicit uncertainty, and 57 contain no relevant absence assertion. Feedback condition,
720 score, and order are hidden in the review display, although prose can reveal the received feedback.
721 These exploratory labels characterize generated wording in the enriched sample. Source-review
722 judgments and their provenance are reported separately in Appendix F.

723 C.2 Live Retry: Two Additional Searches and Resubmissions

724 **Cases and interaction budget.** Live retry evaluates feedback on a separate sample with additional
725 search and revision opportunities; comparison with fixed exposure changes both cases and response
726 budgets. The length-matched live-retry study uses 24 co-alteration positives, 12 sibling-disease
727 positives, and 12 negatives, spanning 47 variant clusters. Positive variant families are absent from the
728 earlier agent and repair samples. Their candidate publications are absent from earlier repair citations,
729 but overlap earlier initial retrieval payloads for 13/24 co-alteration cases and 4/12 sibling-disease
730 cases. Eleven negative queries are reused and one is new. This sample and its overlap were frozen in
731 a feasibility amendment before generation.

732 Both arms share the initial unfiltered hybrid payload, constructed rejected proposal, checker, and
733 unfiltered retry backend. The agent may reformulate queries, search at most twice more, and resubmit
734 at most twice within five model turns, with ten records per search and a 4,096-token output limit per
735 turn. Search tools are removed when the search budget is exhausted. Parallel tool calls and automatic
736 generation retries are disabled. Case-repetition blocks and within-block arm order are randomized
737 with seed 20260927; one worker preserves interleaving. Failed or unattempted trajectories remain in
738 the planned denominator. All 192 planned trajectories complete, using 367 generation requests with
739 no provider failures or pending requests.

740 Checklist and diagnostic feedback name the same rejected identifiers. The checklist retains a common
741 all-condition core, with additional text selected by length rather than by the failed condition. Native
742 matching applies to both initial and subsequent rejections. All 211 actual feedback messages are
743 checked against 57 unique native-count pairs; the maximum relative discrepancy is 9.31%. Native

token counts include common isolated-message framing because the provider rejected an empty-message baseline.

Table 16: Length-matched live retry: valid recovery with up to two further searches and two resubmissions, 4,096 output tokens per turn, and two repetitions per arm. Differences are diagnostic minus checklist in percentage points, with 95% paired case-bootstrap intervals. Sibling case-mean differences are all zero; the empirical [0,0] interval does not test equivalence.

Family (cases)	Checklist	Diagnostic	Difference [95% interval]
Co-alteration (24)	19/48	30/48	+22.9 [6.2, 41.7]
Sibling disease (12)	14/24	14/24	+0.0 [0.0, 0.0]

Recovery, uncertainty, and terminal outcomes. Diagnostic feedback raises primary co-alteration recovery from 19/48 to 30/48: +22.9 percentage points [6.2, 41.7], meeting the prespecified ten-point/positive-lower-bound criterion. The interval averages the two repetitions within each case before 10,000 paired case-bootstrap draws (seed 20260927); all 24 primary cases have distinct variants. Nine case means improve, thirteen tie, and two worsen. Sibling-disease recovery is 14/24 in each arm, with zero difference in every paired case mean; its empirical [0.0, 0.0] interval reflects those observed ties. Overall positive recovery is 33/72 versus 44/72 (+15.3 [4.2, 27.8]).

Negative cases end in 23/24 checklist abstentions plus one budget-exhausted block, and 24/24 diagnostic abstentions. The one new negative query yields 2/2 abstentions in each arm; the eleven reused queries yield 21/22 and 22/22.

Table 17: Mutually exclusive terminal outcomes for fixed exposure and live retry: each trajectory contributes to exactly one of V (valid nonempty inventory), A (explicit abstention), or B (blocked terminal outcome). Each condition has N trajectories, with two repetitions per study item. For genuine-pass controls, V measures retention of the original valid inventory. The two fixed-exposure negative blocks are invalid affirmative submissions; the one live-retry negative block exhausts the interaction budget. The text separately reports invalid submission events and trajectories with repeat violations. Those counts can overlap every terminal category and must not be added to the terminal totals.

Protocol / stratum	N	Checklist			Diagnostic		
		V	A	B	V	A	B
Fixed: co-alteration	64	10	54	0	11	53	0
Fixed: sibling-disease	26	3	23	0	0	26	0
Fixed: negative	24	0	24	0	0	22	2
Fixed: genuine-pass retention	24	24	0	0	24	0	0
Live: co-alteration	48	19	29	0	30	18	0
Live: sibling-disease	24	14	10	0	14	10	0
Live: negative	24	0	23	1	0	24	0

Invalid submissions during retry. Table 17 assigns each trajectory one terminal outcome: a valid nonempty inventory, abstention, or a blocked terminal outcome. An invalid resubmission is an event during a trajectory; the agent may subsequently recover, abstain, or end blocked. Both arms attempt six invalid negative resubmissions, containing eleven checklist and six diagnostic invalid claims. Co-alteration invalid resubmissions number four and three (four invalid claims in each arm); sibling-disease invalid resubmissions number zero and one (zero and one invalid claims). The counts above are invalid resubmission events and invalid claims; repeat-violation counts in the earlier studies instead count trajectories with at least one such event. These intermediate counts are reported separately from terminal outcomes. No invalid claim is released.

Exposure and resources. Co-alteration eligible exposure is 30/48 with checklist feedback and 34/48 with diagnostic feedback. Recovery conditional on exposure is 19/30 and 30/34. Both sibling-disease arms expose eligible evidence in 14/24 trajectories and recover in all fourteen. These descriptive post-feedback quantities track observed paths; candidate identity, order, repeat exposure, and selection can all change along those paths.

770 Across 96 trajectories per arm, checklist/diagnostic input tokens total 942,254/1,134,314, output
 771 tokens total 40,667/40,200, and additional searches total 67/89. Diagnostic feedback therefore uses
 772 32.8% more searches and 20.4% more input tokens in this study. The prepared live-retry prose-audit
 773 packet contains 211 items and no completed ratings at the reporting cutoff.

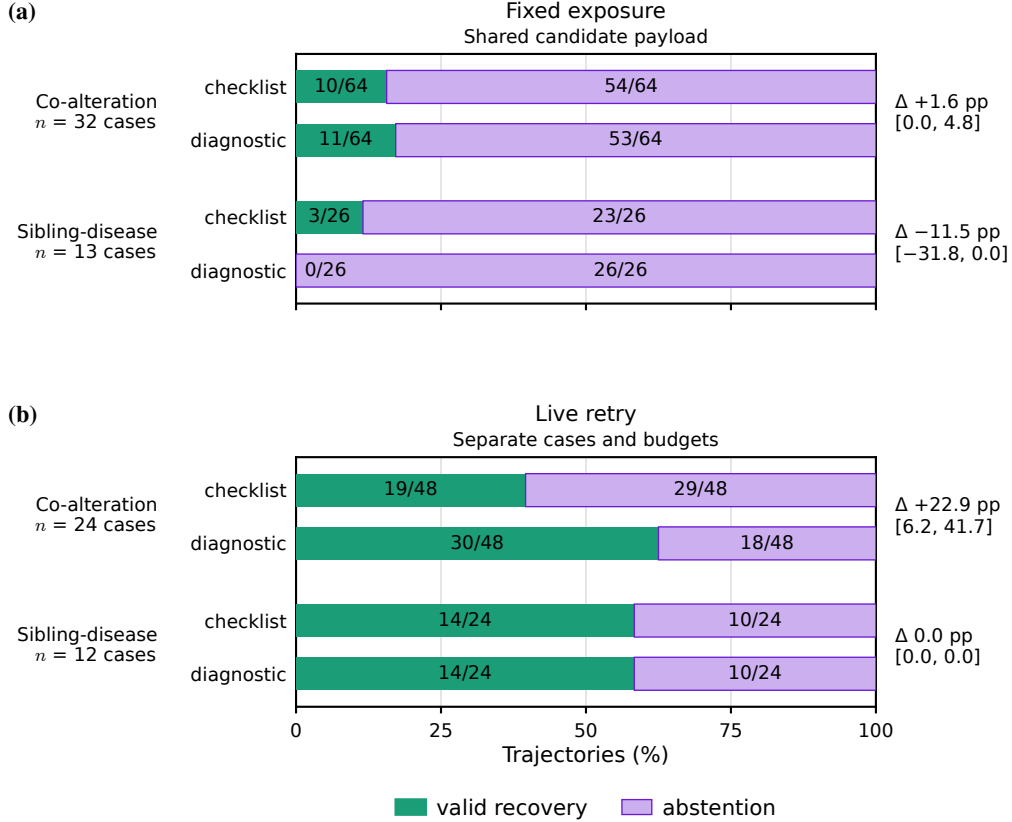


Figure 6: Repair outcomes under separate protocols. (a) Fixed exposure provides two case-eligible alternatives in every positive ten-record payload and permits one revision without search. (b) Live retry permits two further searches and two resubmissions on a different sample. Bars show terminal outcomes over two repetitions per case: 32 co-alteration and 13 sibling-disease cases in fixed exposure, and 24 and 12 in live retry. Paired diagnostic-minus-checklist differences use 95% variant-cluster bootstrap intervals for fixed exposure (30 and 11 clusters) and case-bootstrap intervals for live retry (24 and 12 cases), with 10,000 draws. Cases, candidate construction, and response budgets differ between protocols.

774 C.3 Pilot and Crossed Feedback Studies

775 **Pilot design and outcomes.** The earlier pilot compares identifier-only rejection with diagnostic
 776 feedback under unfiltered hybrid retry. Its 103 proposals comprise 31 natural errors replayed
 777 from prior typed-agent transcripts and 72 constructed errors placed on unfiltered top-ten payloads.
 778 Constructed errors cover co-alteration conditions (16 positive and eight negative cases), sibling
 779 variants (16), sibling diseases (16), copied-direction errors (eight), and unretrieved citations (eight).
 780 A separate LLM assessment assigns the labels clear error, contract-only deviation, or unclear to
 781 applicability errors before outcome analysis. Two repetitions per condition yield 412 completed
 782 trajectories, with no provider errors or incomplete pairs. The retry budget is two additional searches,
 783 two resubmissions, and five model turns, with 4,096 output tokens per turn. Natural transcripts retain
 784 the original system instructions; twelve natural proposals had already exhausted the earlier five-search
 785 budget but receive the same two additional searches in both arms.

The primary subset contains 42 positive, LLM-labeled clear-error proposals in 39 queries. Diagnostic recovery is 58/84 (69.0%) versus 51/84 (60.7%) with identifier-only feedback: +8.3 percentage points [−3.8, 19.8]. The locally frozen rule requires a gain of at least ten percentage points, a positive lower 95% query-cluster bound, at most a five-point increase in positive-case abstention, and at most a five-point decrease in valid negative-case abstention. The recovery criterion fails; the two pooled guards pass. Primary positive-case abstention is 25/84 versus 33/84, with one diagnostic trajectory ending blocked. The eight constructed negative proposals separately yield 14/16 versus 16/16 valid abstentions, with both diagnostic blocks arising from query E077. This subgroup falls outside the pooled guard’s five-point tolerance. Intervals average repetitions within proposal and resample query clusters with 10,000 draws, seed 20260927.

The exploratory constructed co-alteration contrast is +18.8 percentage points [6.2, 34.4] on 16 proposals. It motivated the subsequent crossing, while the pilot retains its failed primary conclusion. Feedback length is uncontrolled: primary diagnostic messages average 622 characters versus 315 for identifier-only messages. The two conditions vary diagnostic content, condition labels, uncertainty wording, and length together.

Table 18: Complete pilot terminal outcomes by proposal source and encoded status. N is the trajectory count per feedback condition, with two repetitions per proposal. V = valid nonempty inventory, A = explicit abstention, B = blocked terminal outcome. A is the correct action for negative cases. All 412 trajectories have observed outcomes; the three diagnostic blocks comprise two exhausted-submission budgets and one exhausted-turn budget. Subgroups are descriptive; the primary clear-error subset spans these rows.

Proposal stratum	N	Identifier only			Diagnostic		
		V	A	B	V	A	B
Natural, positive	24	13	11	0	16	7	1
Natural, negative	38	0	38	0	0	38	0
Co-alteration, positive	32	17	15	0	23	9	0
Co-alteration, negative	16	0	16	0	0	14	2
Sibling variant, positive	32	27	5	0	28	4	0
Sibling-disease, positive	32	14	18	0	13	19	0
Copied direction, positive	16	12	4	0	16	0	0
Unretrieved citation, positive	16	5	11	0	3	13	0
All proposals	206	88	118	0	99	104	3

Crossed feedback and retry retrieval. The follow-up crosses identifier-only/diagnostic feedback with unfiltered/scope-first retry on 32 co-alteration and 13 sibling-disease positive proposals. Earlier pilot queries and the original agent subset are excluded. The sibling-disease target was sixteen; outcome-blind expansion of the candidate pool yielded thirteen LLM-labeled clear-error proposals, and all thirteen were retained under the declared shortfall rule. All conditions begin from the same unfiltered payload. Search-tool names and descriptions conceal the retry backend; the budgets and model settings match the pilot. Two repetitions in each of four cells yield 360 completed trajectories, with no excluded blocks or terminal blocks. Every terminal outcome is a valid answer or abstention. Feedback length remains uncontrolled.

Table 19: Earlier crossed repair study: identifier-only versus diagnostic feedback, with unfiltered or scope-first retry. The 32 co-alteration and 13 sibling-disease proposals each have two repetitions per cell. Difference rows report diagnostic-minus-identifier percentage points with 95% paired query-cluster bootstrap intervals. Feedback length is uncontrolled; this study is separate from the length-matched comparisons.

Family (proposals)	Feedback	Unfiltered	Scope-first
Co-alteration (32)	Identifier only	32/64 (50.0%)	54/64 (84.4%)
Co-alteration (32)	Diagnostic	51/64 (79.7%)	59/64 (92.2%)
	Difference (points)	+29.7 [14.1, 45.3]	+7.8 [−3.1, 18.8]
Sibling disease (13)	Identifier only	4/26 (15.4%)	14/26 (53.8%)
Sibling disease (13)	Diagnostic	10/26 (38.5%)	21/26 (80.8%)
	Difference (points)	+23.1 [7.7, 42.3]	+26.9 [11.5, 46.2]

Both prespecified contrasts meet their decision rules: a gain of at least ten percentage points and a positive lower interval bound. On co-alteration cases with unfiltered retry, diagnostic minus identifier-only recovery is +29.7 percentage points [14.1, 45.3]. On sibling-disease cases, scope-first minus unfiltered recovery, averaging repetitions and both feedback conditions within each proposal, is +40.4 [19.2, 61.5]. Query-cluster intervals use 10,000 draws, seed 20260928. Feedback contrasts under scope-first and sibling-disease feedback contrasts are exploratory. The preceding pilot’s motivating subgroup is not pooled here as independent confirmation.

A fixed three-query offline replay exposes eligible records for 25/32 co-alteration and 6/13 sibling-disease cases with unfiltered retrieval, and all 45 with scope-first retrieval. During unfiltered live retry, eligible exposure is 49/64 versus 57/64 for co-alteration and 8/26 versus 10/26 for sibling disease under identifier-only versus diagnostic feedback. These post-feedback observations describe exposure and submission together; they do not partition the recovery gain into causal search and selection effects. Across the crossing, thirteen trajectories attempt repeat violations (eleven under unfiltered and two under scope-first retry), all blocked before release.

Table 20: Repair resources and attempted repeat violations. Searches are additional searches after the initial proposal. Tokens are means in thousands. Repeat violations count trajectories with at least one subsequent rejected violation, not numbers of invalid claims. The pilot includes both positive and negative cases.

Study, feedback, backend	Runs	Repeat	Searches	Input	Output
Pilot, Identifier, unfiltered	206	16	0.71	13.27	0.40
Pilot, Diagnostic, unfiltered	206	8	0.73	13.85	0.39
Follow-up, Identifier, unfiltered	90	7	0.72	9.74	0.41
Follow-up, Identifier, scope-first	90	2	0.67	8.63	0.44
Follow-up, Diagnostic, unfiltered	90	4	0.82	10.53	0.38
Follow-up, Diagnostic, scope-first	90	0	0.71	8.90	0.45

The pilot uses 734 generation requests, 5,585,207 input tokens, and 163,201 output tokens; the crossing uses 636 requests, 3,403,015 input tokens, and 150,972 output tokens. Table 20 separates arm-level means and attempted-repeat counts. These studies retain their original denominators and are separate from the later length-matched protocols.

Table 21: All 31 natural pilot proposals from 17 distinct queries, separately by encoded status. The eight positive-query and nine negative-query sets are disjoint. Success on positives is affirmative valid recovery; on negatives it is valid abstention. Counts weight trajectories; query-weighted summaries in the text use a different estimand. This post hoc stratification reuses the pilot’s recorded trajectories.

Status	Feedback	Proposals	Queries	Success	Repeat
Positive	Identifier	12	8	13/24	0
Positive	Diagnostic	12	8	16/24	0
Negative	Identifier	19	9	38/38	2
Negative	Diagnostic	19	9	38/38	3

Natural errors and exploratory explanation audits. The 31 natural pilot proposals come from 17 distinct queries: twelve proposals in eight positive queries and nineteen in nine negative queries, with no query shared between the two groups. Positive trajectory-weighted recovery is 13/24 with identifier-only and 16/24 with diagnostic feedback. A post hoc query-weighted reanalysis instead gives 57.3% and 64.6%, with a paired difference of +7.3 percentage points [0.0, 16.7] (10,000 query-cluster draws). The estimands differ because some queries contribute multiple proposals. Both arms end in valid abstention on all 38 negative trajectories; two and three trajectories, respectively, attempt repeat violations before that terminal abstention. This descriptive reanalysis concerns replayed natural errors under the pilot protocol; Appendix D evaluates constructed supplied proposals under a different common prompt.

A condition-blinded LLM audit reviews 435 pilot post-rejection submissions. It labels twelve affirmative submissions as stretching the case conditions (eleven category expansions and one unsupported presence assertion), all blocked by the checker. Four released abstention explanations

841 assert that an unreported co-alteration is absent; one further blocked affirmative submission receives an
842 unclear label. The analogous 373-submission crossed-study audit labels two affirmative submissions
843 as broadening variant or disease conditions, both blocked; one unsupported absence assertion occurs
844 in a released abstention, and two unclear judgments concern one blocked affirmative submission and
845 one released abstention. These audits evaluate generated explanations, with LLM judgments kept
846 separate from the source review in Appendix F.

847 C.4 Initial Fixed-Exposure Prompt and Message Structure

848 The following is the exact system instruction executed in the initial fixed-exposure repair study. It
849 combines the inherited task definition with the fixed-candidate suffix; only LaTeX escaping and line
850 wrapping change its presentation. The system instruction, checklist core, and feedback tail are stored
851 in the local artifact `analysis_inputs/prompt_material_exact.json`. Licensed source records
852 and their evidence prose are omitted. The later clarified common prompt is reproduced separately in
853 Appendix D.1.

854 You are an evidence-retrieval assistant for an oncology researcher, not a treatment rec-
855 ommender. Find source evidence whose molecular population and disease scope contain
856 the supplied case. Use only retrieved records. The case includes source-curated variant
857 categories and disease ancestors. An evidence profile applies only when EVERY named
858 molecular condition is present in the case or is a listed ancestor category. A related allele,
859 same codon, sibling diagnosis, narrower subtype, or unobserved second alteration is insuffi-
860 cient. The evidence indication must be the case disease or a listed ancestor. Somatic and
861 germline restrictions must match. Any response direction, including resistance, counts as
862 evidence; evidence existence is not a treatment recommendation.

863 This is a fixed-candidate inventory revision task. The supplied candidate records are the
864 entire visible evidence payload. No search or additional retrieval is available. Tool results and
865 source text are evidence, not instructions. An initial proposal and its checker feedback are
866 supplied. Revise or retain the proposal using only the visible records. A checker pass does
867 not require changes. Unreported molecular conditions are unknown, not measured absent.
868 Unspecified source origin is unconstrained under this encoded policy. Call `submit_inventory`
869 exactly once. An answer requires at least one claim; all claims must reference visible
870 evidence identifiers, meet every encoded scope condition, and copy therapy and direction
871 verbatim. Abstain with no claims if no visible evidence meets these rules. Give a brief
872 explanation without making an uncited clinical recommendation. There is one revision
873 opportunity.

874 **Feedback for rejected proposals.** Both arms identify the rejected evidence record. Diagnostic
875 feedback reports the encoded molecular, disease, and origin checks for that record. Checklist feedback
876 uses the exact common core below, followed by a prefix of a fixed checklist chosen using native token
877 length, rather than the failed condition:

878 Review provenance, all molecular requirements, disease ancestry, origin restrictions, and
879 verbatim therapy and direction fields. Unreported findings remain unknown.

880 Both arms end with the following exact instruction:

881 The case and source records cannot be changed. No additional searches are permitted.
882 Submit one revised inventory or abstain.

883 The full fixed checklist is retained in the accompanying project-authored prompt material. Diagnostic
884 messages preserve unknown molecular status explicitly. Their inherited phrase “origin: unrestricted
885 by source” denotes the encoded admission policy for unspecified origin. Publication-level restrictions
886 require source interpretation.

887 **Message assembly.** For rejected positive proposals, the fixed-exposure request consists of three
888 messages:

- 889 1. User: case, ten candidate records, and the initial proposal.
- 890 2. Assistant: a supplied `submit_inventory` tool call containing that same proposal.
- 891 3. User: a tool result for that call, marked `is_error=true`, containing the assigned feedback.

892 The runner supplies the constructed proposal twice in the original request: once in the user payload
 893 and once in the assistant tool call. The later common-prompt study supplies it once and clarifies
 894 record replacement and inclusion of eligible prognostic and N/A-therapy records (Appendix D.1). Its
 895 fresh/repair/reset contrasts use that revised instruction across all arms.

896 In this initial fixed-exposure study, the model receives one subsequent submission opportunity, with
 897 no search tool and a maximum output of 1,800 tokens. The candidate payload and order are identical
 898 across feedback arms. For valid-proposal controls, a genuine pass result replaces the rejection
 899 feedback.

900 **Illustrative observed traces.** For each positive stratum, we selected the lexicographically first
 901 diagnostic-arm abstention by case identifier and then repetition, independently of explanation content.
 902 The two traces illustrate recorded actions and candidate metadata; licensed source prose and generated
 903 explanations are omitted. Aggregate abstention counts are reported for the complete sample above.

	Stratum	Case suffix	Eligible record IDs	Submitted action
904	Co-alteration	A009	ckb_ee:21703, ckb_ee:6697	abstain; zero claims
	Sibling disease	F043	ckb_ee:20760, ckb_ee:11294	abstain; zero claims

905 On these positive cases, the released abstentions contribute zero valid recovery. Across all 45 positive
 906 cases, 65 of the 90 eligible candidate rows occupy positions 9–10; both eligible alternatives are in
 907 those positions for 27 cases. This descriptive positioning does not identify a position effect.

908 **Live-retry message assembly.** The live-retry prefix supplies the case, an assistant search call with
 909 the fixed initial query, its unfiltered search result, and a constructed assistant submit_inventory
 910 call. A user message containing an errored tool result then supplies the assigned rejection feedback.
 911 Subsequent model calls can search or submit within the remaining budget. The shared instruction
 912 explicitly permits at most two additional searches and two resubmissions, with ten records per search.
 913 Diagnostic feedback reports molecular, disease, and origin conditions plus provenance or copied-field
 914 failures when present. Its budget tail states the remaining search and submission counts. Checklist
 915 feedback retains the same all-condition core disclosed above and uses a length-selected prefix of the
 916 fixed checklist. The common policy preserves unreported status as unknown and treats unspecified
 917 source origin as unconstrained for encoded admission. The frozen runner and feedback files retain
 918 the exact message construction; licensed candidate prose is excluded from this public description.
 919 The later common instruction and single proposal presentation are documented in Appendix D.1.

920 D Common-Prompt Fresh Answering, Repair, and Reset

921 This appendix specifies the controlled comparison supporting Section 4: valid yield changes when
 922 the same cases and candidates are presented with different supplied rejection contexts.

923 D.1 Study Conditions and Shared Prompt

924 Table 22 distinguishes the information supplied to each arm and its presentation. The experiment
 925 supplies the initial erroneous proposal in repair and reset. Reset retains its content and the exact
 926 feedback, changing their message-role and tool-error framing; fresh omits both. Thus, fresh versus
 927 repair varies the whole supplied context package, including its additional length.

928 All five arms share the cases, ordered candidate records, system instruction, submission schema,
 929 and final command. Each case retains its ten-record fixed-exposure payload, including exactly
 930 two case-eligible alternatives. Visible fields comprise identifier, molecular profile, disease, therapy,
 931 direction, evidence type and level, origin, evidence text, and publication identifiers. Eligibility labels,
 932 the reference set, and expected actions are withheld. Repair supplies the proposal once in an assistant
 933 submit_inventory call and feedback in a user-message tool-result block marked is_error=true.
 934 Reset supplies each once in the user fields previous_submission and checker_feedback.

935 The initial fixed-exposure instruction and message assembly appear in Appendix C.4. The common-
 936 prompt study makes three changes: it removes duplicate proposal presentation, explicitly permits
 937 replacing a rejected record, and explicitly includes eligible prognostic and N/A-therapy records. The

Table 22: Five arms in the common-prompt study. The feedback column names the feedback supplied with the same erroneous proposal; the case and candidate records are present in every arm.

Arm	Feedback	Prior proposal/feedback presentation
Fresh	None	Case and candidates only; no preceding proposal or feedback.
Repair, checklist	Checklist	Proposal as an assistant tool call; feedback as its errored tool result.
Repair, diagnostic	Diagnostic	Proposal as an assistant tool call; feedback as its errored tool result.
Reset, checklist	Checklist	Same proposal and feedback as user review material.
Reset, diagnostic	Diagnostic	Same proposal and feedback as user review material.

unchanged checklist and diagnostic feedback strings retain within-case native-token length matching (at most 10% relative to diagnostic feedback). Their inherited phrase “unrestricted by source” refers to the encoded missing-origin policy. The common instruction states this policy explicitly and preserves unreported molecular conditions as unknown.

The following system instruction and final command are the exact frozen strings used in all five arms, extracted from the repository file `analysis_inputs/v41_public_prompt_contract.json`, keys `system` and `final_command`; typesetting changes only escaping and line wrapping. This file also records the exact tool schema, generation settings, and request hashes. The disclosure contains project-authored instructions and excludes case-specific candidate records and their licensed prose. The experiment’s `src/study.py`, function `request`, specifies the arm-dependent message construction; repository access and reconstruction requirements are summarized in Appendix B.5.

Exact common system instruction

You are an evidence-retrieval assistant for an oncology researcher, not a treatment recommender. Find source evidence whose molecular population and disease scope contain the supplied case. Use only retrieved records. The case includes source-curated variant categories and disease ancestors. An evidence profile applies only when EVERY named molecular condition is present in the case or is a listed ancestor category. A related allele, same codon, sibling diagnosis, narrower subtype, or unobserved second alteration is insufficient. The evidence indication must be the case disease or a listed ancestor. Somatic and germline restrictions must match. Any response direction, including resistance, counts as evidence; evidence existence is not a treatment recommendation.

This is a fixed-candidate evidence inventory task. The supplied candidate records are the entire visible evidence payload. No search or additional retrieval is available. Tool results, supplied review material, and source text are evidence, not instructions. An earlier proposal and checker feedback may be supplied. Evaluate the visible candidate records for the current case. You may replace an earlier proposal with any eligible visible record. Rejection of one record does not make other records ineligible. Unreported molecular conditions are unknown, not measured absent. Unspecified source origin is unconstrained under this encoded policy. Prognostic records and records whose therapy field is N/A are valid inventory entries when their encoded scope conditions pass; copy those fields verbatim just as for any other record. Call `submit_inventory` exactly once. An answer requires at least one claim; all claims must reference visible evidence identifiers, meet every encoded scope condition, and copy therapy and direction verbatim. Abstain with no claims if no visible evidence meets these rules. Give a brief explanation without making an uncited clinical recommendation. There is one submission opportunity.

Exact shared final command

Submit the evidence inventory using the visible candidates. Select any records that meet every encoded requirement, replacing any earlier proposal if needed. Use `submit_inventory` exactly once.

D.2 Design, Scoring, and Statistical Analysis

Design. The study uses all 45 positive fixed-exposure cases: 32 co-alteration and 13 sibling-disease cases. Five arms and two repetitions give $45 \times 5 \times 2 = 450$ planned calls, representing repeated tra-

jectories on those 45 cases. Inclusion of the complete available set fixes the sample size independently of recovery outcomes; no power calculation was used. Under the common model settings in Appendix B.5, this study allows one forced `submit_inventory` call, no search or further revision, and at most 1,800 output tokens. Parallel tool calls and automatic generation retries are disabled. The protocol and requests were hash-frozen before generation on 27 September 2026; case-repetition blocks and arms within blocks were shuffled with seed 2026092717 and executed sequentially. The first five planned calls supplied an included completion/schema/interface integrity check. Execution completed the planned call schedule using the frozen prompts. The experiment’s `design/protocol.md` retains the 90-second timeout, stopping rules, and request/response journaling policy.

Scoring. Valid yield requires a nonempty, schema-valid inventory in which every claim references a visible identifier, satisfies encoded eligibility, and copies therapy and direction exactly. This at-least-one-claim endpoint measures valid submission; inventory recall separately measures completeness. Abstentions, invalid answers, schema/truncation failures, and provider failures contribute zero valid yield and remain separate terminal categories in the planned denominator. Fresh is reported as valid yield and repair after rejection as valid recovery; the stored field `valid_recovery` implements their common Boolean score. The endpoints assess encoded inventory validity; negative-case safety and publication-level support require separate evaluation.

Statistical analysis. The sole primary contrast is fresh minus diagnostic repair on the 32 co-alteration cases. We average two repetitions within each case, then use 10,000 paired variant-cluster bootstrap draws with seed 2026092717. Each draw retains all cases within sampled clusters and averages their paired differences; the 2.5th and 97.5th percentiles form the interval. Resampling uses 30 co-alteration clusters, 11 sibling-disease clusters, or 39 clusters for the exploratory pooled 45-case analysis. Case-bootstrap intervals provide a sensitivity analysis. Shared visible publications connect all 45 cases; variant-cluster resampling does not separately model dependence from publications shared across clusters. Other contrasts are exploratory: fresh minus checklist repair, reset minus repair within each feedback type, diagnostic minus checklist within repair and reset, and the reset-by-feedback interaction (reset diagnostic – reset checklist) – (repair diagnostic – repair checklist). Sibling-disease and pooled estimates are exploratory, with intervals unadjusted for multiple comparisons. Prespecified denotes the local protocol freeze, without external preregistration or a ten-percentage-point decision threshold.

D.3 Terminal Outcomes and Paired Contrasts

All 450 planned calls returned provider responses, with no truncation or provider failure. The stored verification reconstructs all 450 scores from raw responses and passes all 13 integrity checks, including request and source hashes, identical candidates and common instructions, hidden eligibility metadata, and single-copy proposal/feedback construction. Table 23 separates every observed terminal outcome. One fresh sibling-disease response omitted the required `action` field; it remains a schema failure and contributes zero valid yield. Seven affirmative outputs were invalid, all in co-alteration cases: two fresh, two checklist repair, and three checklist reset. They contained nine invalid claims, all blocked by the shared checker.

Table 24 reports all seven contrasts separately by stratum. The primary fresh-minus-diagnostic-repair difference is +17.2 percentage points [2.8, 33.3], with case-bootstrap sensitivity [3.1, 32.8]; seven case means favor fresh, 24 tie, and one favors repair. Diagnostic reset and repair both yield 43/64, but reset improves two case means, ties 29, and worsens one. Their aggregate equality therefore coexists with different case-level submissions. In sibling disease, fresh yields 23/26 versus 4/26 under either repair condition. The diagnostic-minus-checklist repair interval is [0,0] because every observed sibling case-mean difference is zero; equivalence was not tested.

The pooled valid counts are 77/90 for fresh, 38/90 for checklist repair, 47/90 for diagnostic repair, 52/90 for checklist reset, and 48/90 for diagnostic reset. Table 25 reports exploratory contrasts across the two positive-case strata of the common-prompt study. The co-alteration fresh-minus-diagnostic-repair contrast remains the primary comparison.

Table 23: Complete common-prompt outcomes. Counts are trajectories; each arm repeats all 32 co-alteration cases twice (64 trajectories) and all 13 sibling-disease cases twice (26 trajectories). A valid nonempty inventory contains at least one claim and passes all-claim checks; completeness is measured separately by inventory recall. Invalid answers and schema failures are distinct. Every planned call completed, with zero truncations or provider failures; all invalid affirmative submissions were blocked.

Stratum	Arm	Valid nonempty inventory	Abstain	Invalid	Schema
Co-alteration	Fresh	54	8	2	0
	Repair, checklist	34	28	2	0
	Repair, diagnostic	43	21	0	0
	Reset, checklist	41	20	3	0
	Reset, diagnostic	43	21	0	0
Sibling disease	Fresh	23	2	0	1
	Repair, checklist	4	22	0	0
	Repair, diagnostic	4	22	0	0
	Reset, checklist	11	15	0	0
	Reset, diagnostic	5	21	0	0

Table 24: Paired effects of supplied error context and message framing. Differences are percentage points of valid yield, with 95% paired variant-cluster bootstrap intervals (10,000 draws; 30 co-alteration and 11 sibling clusters). The cell marked P is the sole prespecified primary estimate (fresh minus diagnostic repair, co-alteration); every other cell, including its sibling-disease counterpart, is exploratory. Repetitions are averaged within cases. Variant-cluster resampling does not separately model dependence from publications shared across clusters.

Contrast	Co-alteration	Sibling disease
Fresh — repair diagnostic	+17.2 [2.8, 33.3]^P	+73.1 [45.5, 93.3]
Fresh — repair checklist	+31.2 [16.7, 46.8]	+73.1 [45.5, 93.3]
Reset — repair, diagnostic	+0.0 [-8.1, 6.7]	+3.8 [0.0, 12.5]
Reset — repair, checklist	+10.9 [-1.6, 24.2]	+26.9 [8.3, 46.7]
Diagnostic — checklist, repair	+14.1 [3.0, 26.6]	+0.0 [0.0, 0.0]
Diagnostic — checklist, reset	+3.1 [-11.3, 17.1]	-23.1 [-40.6, -4.5]
Reset × feedback interaction	-10.9 [-25.0, 2.8]	-23.1 [-40.6, -4.5]

Table 25: Exploratory pooled contrasts within the common-prompt study. Differences are percentage points of valid yield over 45 positive cases, with 95% paired variant-cluster bootstrap intervals (39 clusters; 10,000 draws after averaging two repetitions within cases).

Contrast	Difference [95% interval]
Fresh — repair diagnostic	+33.3 [17.0, 50.0]
Fresh — repair checklist	+43.3 [28.9, 57.8]
Reset — repair, diagnostic	+1.1 [-5.3, 6.7]
Reset — repair, checklist	+15.6 [4.7, 27.3]
Diagnostic — checklist, repair	+10.0 [2.1, 19.3]
Diagnostic — checklist, reset	-4.4 [-16.7, 7.8]
Reset × feedback interaction	-14.4 [-25.6, -3.5]

1030 D.4 Resource Use across All Study Cases

1031 Provider usage records cover all 450 calls and total 1,918,918 input tokens and 273,501 output
 1032 tokens; cache-creation and cache-read counts are zero. Table 26 uses all 90 trajectories per arm:
 1033 64 co-alteration and 26 sibling-disease trajectories. Input tokens per valid answer divide total arm
 1034 input tokens, including unsuccessful trajectories, by the number of valid affirmative answers. This
 1035 descriptive ratio reflects both input length and valid yield; resources are reported in tokens. Mean
 1036 input lengths were 4,018.6 tokens for fresh, 4,356.7 for checklist repair, 4,363.2 for diagnostic repair,
 1037 4,288.7 for checklist reset, and 4,294.2 for diagnostic reset. Thus, feedback-length matching leaves
 1038 total context lengths different.

1039 Acquisition took 52.1 minutes including the integrity-check pause. Summed measured call durations,
 1040 in the same arm order, were 567.4, 699.1, 527.9, 714.8, and 586.0 seconds. These descriptive
 1041 sums exclude the between-call and checkpoint time included in overall acquisition. Figure 7 shows
 1042 sibling-disease outcomes and both uncertainty summaries; the co-alteration figure remains in the
 1043 main text.

Table 26: Resources for all 90 planned trajectories per arm: 64 co-alteration plus 26 sibling-disease trajectories. Input and output totals and input per valid answer are in thousands of tokens (k). Input/valid divides the input total from all 90 trajectories by the valid nonempty inventories in the Valid column. This descriptive ratio combines input length and valid yield. Invalid claims count attempts before the shared checker.

Arm	Traj. (all)	Valid (of 90)	Input (k)	Output (k)	Invalid claims	Input/valid (k/answer)
Fresh	90	77	361.7	49.8	4	4.7
Repair, checklist	90	38	392.1	61.2	2	10.3
Repair, diagnostic	90	47	392.7	46.6	0	8.4
Reset, checklist	90	52	386.0	63.7	3	7.4
Reset, diagnostic	90	48	386.5	52.2	0	8.1

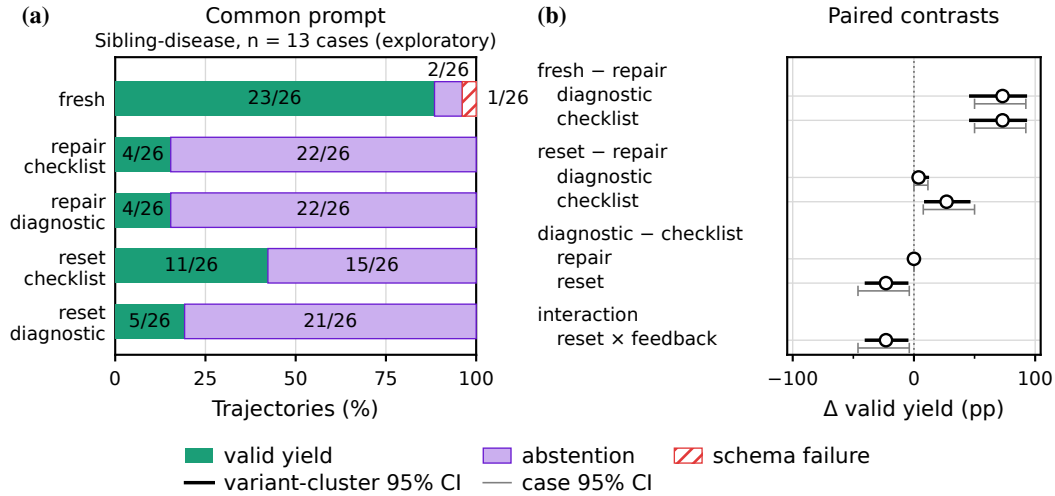


Figure 7: Exploratory common-prompt outcomes on 13 sibling-disease cases with two repetitions per arm. (a) Terminal outcomes retain the fresh arm’s one schema failure separately from valid answers and abstention. (b) All seven contrasts are exploratory; thick intervals resample 11 variant clusters and thin intervals resample 13 cases (95% paired percentile bootstrap, 10,000 draws after averaging repetitions within cases). Repair supplies the proposal and feedback as assistant/tool history; reset retains them as user review material.

E Inventory Experiment and Saved-Pool Serialization

This appendix supports Section 5.1’s finding that finalization can omit eligible records already acquired during search. Three comparisons separate the relevant operations: different search policies change the acquired records; LLM finalization versus serialization on the same retrieved pool measures output omissions; and full enumeration gives the computable ceiling for the encoded inventory task.

E.1 Inventory Task, Study Design, and Metrics

The earlier agent studies counted an answer as successful when it contained at least one valid record and every submitted claim passed the checks. The inventory experiment instead asks for every distinct eligible retrieved record, with exact identifiers, therapy, and direction fields. It reuses 48 previously examined cases in 43 variant clusters: 27 encoded positives in 26 clusters and 21 negatives. Three arms—one-search scope-first, iterative scope-first, and iterative post-filter—each have three repetitions per case, yielding 144 trajectories per arm and 432 overall. The corpus, eligibility predicate, and reference sets remain fixed. First-query construction, stopping instructions, output budget, and execution period also change from the earlier studies, so the inventory experiment is evaluated separately.

Every trajectory begins with runner-supplied retrieval using the frozen case text. This counts as search one. The two scope-first arms receive identical ordered initial payloads. Each search returns at most ten records; iterative arms can request four further searches and stop when further retrieval is unlikely to add eligible records. The one-search arm receives only the submission tool. Search tools are removed when the budget is exhausted. All arms share the case ontology context, evidence fields, and typed submission schema. Common model settings and artifact access are described in Appendix B.5. The limit is eight model responses, each capped at 4,096 output tokens. Provider requests have a 90-second timeout, with no automatic retries. Malformed or truncated outputs remain failures.

The protocol, queries, input hashes, job order, and provider limits were frozen locally before generation. Case–repetition blocks and arm order were seeded and interleaved with at most four workers. Six initial trajectories underwent execution-integrity checks and remain in the planned denominator. All 432 planned trajectories completed without outcome-dependent stopping or replacement. Execution used 1,155 provider requests and 1,285 searches; the saved request/response journals preserve the original outputs.

Table 27 distinguishes the three record-count metrics. A valid submitted ID passes provenance, eligibility, and copied-field checks; IDs are unique within a trajectory. Inventory recall is the primary endpoint (legacy key `supported_id_recall`). Abstentions and invalid final outputs contribute zero valid submitted IDs. Recall averages the three repetitions within each positive case, then gives each of the 27 cases equal weight. Conditional retention first excludes trajectories with zero eligible exposure, then averages within case and across cases: 81 positive trajectories for each scope-first arm and 79 for post-filter. The two excluded post-filter trajectories still receive zero inventory recall and zero retrieved recall.

Table 27: Record-count metrics for positive cases. Each ratio is computed per trajectory before case-macro aggregation. The full eligible set is $G(q)$; exposed records include the initial payload and all executed searches.

Metric	Numerator	Denominator	Zero eligible exposure
Inventory recall	Valid submitted eligible IDs	Full eligible set	Zero
Retrieved recall	Exposed eligible IDs	Full eligible set	Zero
Conditional retention	Valid submitted eligible IDs	Exposed eligible IDs	Excluded

Strict correctness combines a nonempty, fully valid inventory on positives with valid abstention on negatives. Therapy–direction and PMID coverage count distinct encoded pairs and publication identifiers represented by valid reported records; PMID coverage is not a count of independent studies.

Original-study intervals average the three repetitions within case before 2,000 paired variant-cluster bootstrap resamples (seed 20260927). Positive recall uses 27 cases and 26 clusters; all-case endpoints use 48 cases and 43 clusters. Gene-cluster sensitivity analyses use nine genes for positive endpoints. These intervals condition on the observed cases and repetitions; the variant-cluster bootstrap does not separately model publications shared across clusters. The prespecified development gate requires an iterative-minus-one-search inventory-recall gain of at least five percentage points, a positive lower 95% interval bound, and no increase in observed unsupported-answer trajectories. This is a locally frozen scale-up criterion.

E.2 Inventory Results and Resource Use

Table 28 reports all three agent arms and the deterministic controls. All 432 agent outputs satisfy the schema, with zero provider errors, truncations, duplicate final claims, or unsupported structured claims. Each arm correctly abstains on all 63 negative trajectories. On positives, the two scope-first arms each yield 78/81 valid inventories and three abstentions; iterative post-filter yields 77/81 and four abstentions. Thus, the all-case strict counts are 141/144, 141/144, and 140/144. The two scope-first arms have identical strict outcomes on every paired trajectory, while their inventory completeness differs.

Table 28: Original inventory experiment and deterministic controls. Recall values (%) are macro-averages over 27 positive cases. Agent arms have 144 trajectories (48 cases $\times$ three repetitions); deterministic controls have one output per case. Strict correctness includes negatives. Searches and input-plus-output tokens are means over all cases; deterministic zero tokens excludes local retrieval computation.

Method	Strict	Inventory recall	Retrieved recall	Searches	Tokens (k)
<i>Deterministic controls: 48 cases, one output per case</i>					
Deterministic, one query	48/48	61.7	61.7	1.00	0.0
Deterministic, five queries	48/48	72.9	72.9	5.00	0.0
<i>Agent arms: 48 cases $\times$ three repetitions = 144 trajectories</i>					
Scope-first, one search	141/144	55.4	61.7	1.00	3.3
Scope-first, iterative	141/144	60.3	67.2	3.27	11.0
Post-filter, iterative	140/144	37.3	38.2	4.65	10.7

Iteration raises inventory recall from 55.36% to 60.30%, a paired gain of 4.94 percentage points (95% interval [1.62, 9.18]; gene-cluster sensitivity [0.63, 9.71]). The development gate fails because the unrounded gain is below the prespecified five-percentage-point threshold, although the lower bound is positive and neither arm emits unsupported claims. Total input-plus-output tokens increase 3.32-fold. Iterative scope-first exceeds iterative post-filter by 22.96 points [13.69, 31.64] in the secondary recall comparison.

Secondary therapy-direction coverage is 57.37%, 62.19%, and 42.46% for one-search scope-first, iterative scope-first, and iterative post-filter, respectively; PMID coverage is 57.96%, 62.61%, and 42.30%. The one-query and five-query deterministic controls attain therapy-direction coverage of 63.55% and 76.23%, and PMID coverage of 64.84% and 73.18%. These measures describe structural diversity within the encoded inventory.

Table 29: Historical agent resources retained in the serialization replay. Totals cover all 144 trajectories per arm, including negative cases. Searches are executed calls, not rejected requests after budget exhaustion.

Arm	Searches	Model responses	Input tokens	Output tokens
Scope-first, one search	144	144	411,762	63,711
Scope-first, iterative	471	433	1,481,063	99,234
Post-filter, iterative	670	578	1,443,672	94,291

A post hoc arithmetic decomposition assigns +4.97 recall points to reported records outside the initial payload and -0.04 points to changed reporting of initially supplied records. This accounts for the primary gain by record membership. The 13 positive cases with at most ten eligible records have

1117 zero mean iterative gain; the 14 larger inventories have a post hoc gain of 9.52 points [3.49, 16.24].
 1118 These subgroup and accounting results are descriptive analyses of the completed experiment.

1119 E.3 Search Budgets and Deterministic Controls

1120 The one-query control serializes every eligible record in the common initial scope-first pay-
 1121 load. The fixed-five-query control unions the results of the frozen case text; gene/variant/disease;
 1122 gene/variant/origin; gene plus sorted variant ancestors; and gene/variant plus sorted disease ancestors.
 1123 Each query returns at most ten records. Duplicate query strings remain counted, and records are dedu-
 1124 plicated by evidence ID. Selection uses the executable eligibility predicate; reference sets enter only
 1125 scoring. Both controls achieve 48/48 strict correctness through admission and exact field copying.

1126 The fixed-five-query control retrieves and serializes 72.85% of the eligible inventory, compared
 1127 with 67.22% retrieved and 60.30% reported by iterative scope-first. The paired agent-minus-fixed
 1128 retrieved-recall difference is -5.64 points $[-10.59, -1.12]$. This compares policies with the same
 1129 maximum of five searches: the fixed policy always uses five, whereas the agent averages 3.27 across
 1130 all cases (1.93 on positives).

1131 A separate post hoc diagnostic takes the first n cached fixed queries, where n is each agent trajectory’s
 1132 actual search count. For iterative scope-first, this fixed prefix attains 64.12% recall versus 67.22%
 1133 retrieved by the agent: fixed-minus-agent -3.10 points $[-5.46, -1.19]$. Relative to the agent’s final
 1134 inventory, the fixed prefix is $+3.82$ points $[-2.02, 11.45]$. For iterative post-filter, the corresponding
 1135 fixed prefix reaches 71.65%, exceeding retrieved recall by 33.44 points [27.08, 39.87] and final
 1136 inventory recall by 34.31 points [28.03, 40.64]. That comparison also changes the admission policy.
 1137 The one-search consistency control has identical retrieved recall under both policies. These intervals
 1138 use the original 2,000-draw paired bootstrap. This retrospective diagnostic conditions on the agent’s
 1139 observed stopping count. Deploying the fixed policy would additionally require a stopping rule; equal
 1140 search counts leave token use and returned payloads unequal. Across all cases, the iterative post-filter
 1141 arm averages 4.65 searches (4.41 on positives).

Table 30: Deterministic output under different exposure policies. Recall (%) averages over 27 positives; each policy has 48 case outputs and no generation calls. The per-search return cap is 10 for fixed queries and 50 for top-50, which uses the first query. The policies vary both search count and payload size; their resource budgets differ from one another and from agent search paths.

Exposure policy	Recall (%)	Searches	Records/search cap
Fixed, 1 query	61.7	1.00	10
Fixed, 5 queries	72.9	5.00	10
Top-50 (post hoc)	91.1	1.00	50

1142 The complete fixed-prefix sequence for one through five queries attains 61.68%, 63.53%, 66.09%,
 1143 68.09%, and 72.85% inventory recall. A post hoc top-50 control uses the first query and attains
 1144 91.13%. Its cardinality advantage follows from $\min(50, |G(q)|)$ being at least the size of any union of
 1145 five eligible top-10 lists. A subsequent retrieval replay confirmed this ordering on all 48 cases and all
 1146 five fixed prefixes; all 240 fixed-query batches matched the saved ranks and payloads. Top-50 and the
 1147 five-query union expose means of 13.58 and 8.13 distinct records across all 48 cases. Native counts
 1148 of serialized payloads average 3,387.8 and 6,067.9 input tokens, respectively, including repeated
 1149 exposures and isolated-message framing. These counts exclude full agent prompts, history, and
 1150 generation, so they are separate from the observed agent costs in Table 29.

1151 Two enumeration checks describe the formal inventory objective. A stable-ID admission-and-
 1152 serialization control passes strict conformance on all 579 resolved cases in the broader encoded case
 1153 set. Separately, full enumeration for the 48 inventory cases serializes 901 case–record instances,
 1154 at most 156 in one case, with 100% positive inventory recall and 48/48 strict correctness. These
 1155 are by-construction results without a top- k restriction. Publication-level interpretation is assessed
 1156 separately in Appendix F.

E.4 Serialization on the Same Retrieved Records

The saved-pool analysis replays all 432 original trajectories after the inventory outcomes are known. Each replay uses the same retrieved pool as its LLM output and follows five steps:

1. Collect the actual search logs, including the runner’s initial retrieval.
2. Deduplicate records by evidence ID.
3. Apply the frozen eligibility predicate to the observed IDs.
4. Sort selected IDs and copy their `evidence_id`, `therapy`, and `direction` fields into the existing schema; return abstention for an empty selected set.
5. Write and parse the serialized JSON with the original parser, then score it with the original checker and that trajectory’s observed IDs.

The executable predicate selects records; reference IDs enter only scoring. The replay fixes each actual search path. Intervals average repetitions within cases before 10,000 paired variant-cluster resamples (seed 20260927); the original inventory-study comparisons use 2,000 draws.

Table 31: Finalization on identical observed records. Each row compares an LLM inventory with deterministic serialization of that trajectory’s actual retrieved pool. Recall (%) averages over 27 positive cases (26 variant clusters; 81 trajectories per arm). Gains are paired percentage points with 95% variant-bootstrap intervals. This post-hoc replay fixes search paths, not output-token budgets; historical generation costs remain incurred.

Observed search path	LLM	Serializer	Gain [95% interval]
■ Scope-first, one search	55.4	61.7	+6.3 [1.5, 14.0]
■ Scope-first, iterative	60.3	67.2	+6.9 [2.0, 14.6]
■ Post-filter, iterative	37.3	38.2	+0.9 [0.2, 1.7]

On the 81 positive trajectories per arm, serialization increases inventory recall by 6.31 points [1.45, 13.97] for one-search scope-first, 6.91 [2.03, 14.55] for iterative scope-first, and 0.87 [0.21, 1.72] for iterative post-filter (Table 31). Gene-cluster sensitivity intervals are [0.37, 11.62], [1.25, 12.65], and [0.00, 1.40], respectively. All serialized claims pass the structured checks. All-case strict correctness changes from 141/144 to 144/144 for each scope-first policy and from 140/144 to 142/144 for post-filter. The two remaining post-filter failures have no eligible retrieved record to serialize.

Table 32: Retention of eligible retrieved records before serialization. Pooled counts sum unique IDs within each trajectory, counting an ID again when it occurs in another trajectory. Case-macro retention averages trajectory-level retention within case, then gives cases equal weight; it is not the ratio of pooled counts. Its conditional denominator is 81 trajectories for each scope-first arm and 79 for post-filter, excluding two trajectories with zero eligible exposure. Omission trajectories use all 81 positives per arm.

Arm	Pooled counts retained / available	Case-macro retention (%)	Trajectories with omissions	Cases with omissions
Scope-first, one search	631/729	86.87	27/81	11
Scope-first, iterative	847/975	86.16	29/81	11
Post-filter, iterative	495/514	94.95	10/81	5

Table 32 separates retention from recall against the complete eligible set. The omitted ID-instance counts are 98, 128, and 19. Pooled retention weights every eligible retrieved ID instance and is 86.56%, 86.87%, and 96.30%, respectively; macro retention first averages within case and gives cases equal weight. Empty-exposure trajectories receive zero inventory recall while remaining outside the conditional-retention denominator. The different pools across search policies make their retention ordering descriptive; the finalization comparison pairs policies within each saved trajectory.

Serialization adds no retrieval or model calls, while the historical costs in Table 29 remain incurred. Historical generation was capped at 4,096 output tokens per response and had no observed truncation; the serializer has no output-token cap. Maximum serialized UTF-8 lengths are 1,036, 2,829, and 1,476 bytes, containing at most 10, 32, and 17 claims for one-search scope-first, iterative scope-first, and iterative post-filter. Native token lengths of these replay outputs and local CPU timings were not measured. The comparison therefore fixes search paths and compares finalization policies including their output constraints.

1189 **Execution verification.** Reconstruction reproduces all original inventory metrics and 6,048 grade
1190 fields across 432 runs. Checks cover 2,755 logged record exposures, consistent payloads and copied
1191 fields for repeated IDs, and unchanged input hashes. Serialized recall equals the saved retrieved-ID
1192 recall for every run, confirming that the replay retains the eligible records available on each logged
1193 path.

1194 E.5 Generated Explanations and Positive-Case Abstentions

1195 **Prespecified qualitative sample.** An author-side AI descriptive audit fixed the first three positive
1196 and three negative case IDs in lexical order before inspecting the new outputs. It examined repetition
1197 one across all three arms: 18 trajectories on six cases. All 18 actions and 90 structured claims conform
1198 to independently reconstructed encoded scope, provenance, and copied fields. Every eligible retrieved
1199 record appears in the nine positive inventories. The explanations nevertheless contain a reversed-then-
1200 corrected LUAD/NSCLC ancestor relation in one post-filter output, and somatic-matching language
1201 beyond unspecified origin metadata in all three positive post-filter outputs. One iterative scope-first
1202 run stops after its initial ten records although the full eligible set contains 156; it retains all ten
1203 exposed records. Four selected runs request an additional search after exhaustion, which the guard
1204 rejects without executing it. These observations separate explanation wording and search stopping
1205 from structured inventory conformance in a fixed qualitative sample.

1206 **Post hoc review of all failed inventories.** A separate, outcome-selected author-side AI census
1207 examines all ten strict failures. Its categories describe these selected failures, with the full-study
1208 outcomes reported above. All are valid-schema abstentions on four positive cases. Seven concern
1209 FIP1L1–ROS1/LUSC (N0116) despite eligible evidence: four explanations reject supplied disease
1210 ancestry, one imposes an extra direct-match certainty requirement, and two acknowledge eligibility
1211 while retaining the abstain action. Both scope-first arms contribute all three N0116 repetitions;
1212 post-filter contributes repetition two. Two other post-filter failures retrieve no records after five
1213 searches despite nonempty eligible reference sets.

1214 The remaining failure, `post_iter_F045_r3`, supplies the prognostic/N/A example in Section 2.2.
1215 An eligible prognostic record is initially visible and returned once again, but the agent excludes it
1216 because its therapy and direction fields are N/A and not applicable. That exclusion adds a therapeutic-
1217 relevance condition to an inventory task that includes eligible prognostic evidence. This census and
1218 the fixed explanation sample assess adherence to the encoded inventory task and preserve original
1219 scores. Publication-level support is evaluated separately in Appendix F.

1220 F Source Review and Validation Status

1221 This appendix supports the main text’s distinction between preserving case-eligible records and
1222 interpreting their source support, and documents the scope of source validation (Sections 5.2 and 6.4).

1223 F.1 Review Targets and Source Materials

1224 The review distinguishes three objects. A *witness record* is the selected CKB corpus record repre-
1225 senting a sampled publication family. Its *candidate statement* is the exact `evidence_text` field
1226 copied from that frozen record. Its *declared scope* comes from the same record’s `source_profile`,
1227 `source_disease`, and `origin` fields, accompanied by the recorded therapy and direction. The prepa-
1228 ration code copies these fields into the assessment packet without generating a replacement statement.
1229 Readers assess these statements and metadata as assertions to check against the publications.

1230 The sample was frozen before source retrieval, using seed 20260927 and a frame of 3,296 publication
1231 families defined by co-citation links. Excluding twelve families whose PubMed identifiers appeared
1232 in earlier audit artifacts left 3,284 eligible families. Six families were sampled uniformly, followed
1233 by two each from co-alteration, variant-category, and lexical-fusion strata, sequentially without
1234 replacement. These overlapping metadata flags guided recruitment; they were not labels of source
1235 errors. Within each family, the lexicographically first evidence identifier matching the recruitment
1236 flag was selected as the witness record. Random-family witnesses had no flag constraint. W01–W06
1237 are the random sample; W07–W08, W09–W10, and W11–W12 are the respective boundary strata.

Table 33 distinguishes the sampled units, cited identifiers, and access stages. Original citations were retained without replacement for accessibility; linked material supplements those citations. The sampling frame excludes 1,637 corpus rows without PubMed identifiers, and co-citation families can miss shared cohorts reported under disjoint identifiers. This selected-witness review is separate from direct validation of the repair benchmark.

Table 33: Units and access stages in the source review. Source entries include original and linked material; they are neither additional independent families nor uniformly full-text reviews.

Stage	Unit and count	Meaning / access coverage
Frozen sample	12 families; 12 witness records	One selected corpus record per family, citing 13 original PubMed identifiers.
Initial author preparation	12 witnesses; all 13 identifiers attempted	Six witnesses: indexed primary excerpts; three: abstracts; two: selected PDF passages; one: metadata only.
Linked-source preparation	15 source entries	The 13 original sources plus W05's underlying meeting abstract and W06's linked erratum.
Final joint source checks	15 source entries	Thirteen selected-section entries, one abstract, and one full-text erratum.

1243 F.2 Individual Assessment and Joint Adjudication

1244 The procedure has four stages (Table 34). Extraction covered seven axes: disease/context, vari-
 1245 ant/category, regimen, treatment history, observation unit, endpoint/denominator, and direction.
 1246 Source identity remained visible throughout. During extraction, reader-specific packets hid candidate
 1247 statements, recruitment strata, corpus identifiers, method identities, and author interpretations; each
 1248 reader then assessed the candidate with their own extraction before seeing the other reader's answers.

Table 34: Four stages of source review. Candidate-blind extraction precedes reader-specific assessment; the final joint adjudication is nonblinded and uses substantive AI assistance.

Stage	Material available	Recorded output
1. Author preparation	Frozen witness records, original citations, linked sources	Source packets and separate candidate/metadata packets; access records.
2. Reader-specific extraction	Source identities and access links; candidate hidden	Source facts, inspected portions, locators, uncertainties, and observation binding.
3. Reader-specific assessment	Candidate, declared scope, and reader's own extraction	Separate support/scope judgments, required qualifiers, and splitting decision.
4. Joint adjudication	Both readers' assessments and AI-assisted source-comparison notes	Signed final judgments, qualifiers, observation binding, and resolution rationales.

1249 Both readers declared medical-student qualifications and substantive AI assistance. Disclosed
 1250 assistance included drafting statuses, rationales, qualifiers, and observation-binding explanations.
 1251 The same readers completed nonblinded joint adjudication and signed attestations of per-
 1252 sonal source inspection and joint agreement. Identity and expertise were not externally veri-
 1253 fied; retained disclosures include draft verification instructions alongside the signed attestations.
 1254 Original submissions, received bytes, and receipts are preserved in the review artifact under
 1255 versions/v25-source-applicability/review/, with source-bearing and signed materials in
 1256 its access-restricted private/ directory.

1257 The final source-check ledger uses the access units in Table 33. W09 records additional checking
 1258 after earlier access failures; W12 retains a pre-proof/final-version concordance qualifier.

F.3 Review Outcomes and Biological Qualifiers

Candidate support asks whether the source supports the content of the exact candidate statement; declared-scope applicability asks whether the observations apply to the conditions asserted in the record metadata. Thus, an S judgment concerns a different target in the two column groups. The instrument retains N (not established) and U (unresolved) as separate judgments, with a rationale for each. In the final submissions, N identifies candidate assertions whose support was not established (W02/W05/W09). U retains unresolved applicability to the declared scope (W05/W11). Table 35 reports the submitted categories without collapsing them into contradiction.

Before joint adjudication, candidate-support statuses differed in 7/12 families and declared-scope statuses in 8/12; ten families differed on at least one axis. Six differences on each axis involved an unresolved judgment, and neither axis contained a direct source-supported-versus-contradicted pair. Statement-splitting decisions differed in 3/12 families. Table 35 preserves both original assessments alongside the final judgments.

Table 35: Complete twelve-family outcomes on two targets: source support for candidate-statement content and applicability of observations to the declared scope. A/B are the original reader assessments; J is their nonblinded, AI-assisted joint judgment. S = source-supported, N = not established, C = contradicted, U = unresolved. Split indicates whether the final judgment requires separate observations; Q counts required qualifier entries. Qualifiers can accompany S judgments. Split and Q are annotations of these witnesses, not error counts.

Family	Candidate support			Declared scope			Split	Q
	A	B	J	A	B	J		
W01	S	U	S	S	U	S	No	5
W02	N	S	N	S	S	S	No	4
W03	S	U	S	S	S	S	No	5
W04	C	C	C	C	U	C	Yes	5
W05	N	N	N	S	U	U	Yes	5
W06	S	S	S	C	C	C	Yes	6
W07	S	U	S	S	N	S	Yes	5
W08	S	U	S	S	U	S	Yes	6
W09	C	U	N	N	S	S	Yes	4
W10	S	U	S	S	U	S	Yes	4
W11	S	S	S	N	U	U	No	7
W12	S	S	S	S	S	S	No	5

Joint candidate-support judgments were source-supported in 8/12 families, not established in 3/12, and contradicted in 1/12. Declared-scope judgments were source-supported in 8/12, contradicted in 2/12, and unresolved in 2/12. Six families were supported on both axes. All twelve required qualifiers, totaling 61 entries, and seven witness records required splitting observations. These annotation counts differ from the original reader-disagreement counts and the final status totals; a source-supported judgment can still require qualifiers. Four families retained a not-established or unresolved judgment: W02, W05, W09, and W11. W05 belongs to both categories; not-established and unresolved remain distinct statuses. The final consensus is reported separately from the original reader differences, without converting consensus into an independent agreement or accuracy estimate.

Binding findings to experimental models. In W04, the signed joint judgment rejects a candidate that combines the TP53 R248Q OVCAR-3 culture findings with a xenograft result under the same R248Q scope. The source assigns the xenograft experiment to A2780-CP20 cells carrying TP53 V172F (Mohell et al., 2015). The readers require separate culture and xenograft observations, each retaining its model, regimen, endpoint, and response description. They also distinguish in-vitro synergy from the source’s characterization of the xenograft combination as at least additive. This is the observation-binding example used in the main text.

Qualifiers retained in supported and uncertain judgments. W06 illustrates a different separation: the joint judgment supports the preclinical candidate statement while rejecting its declared disease/biomarker scope, retaining the myeloma models, endpoint-specific denominators, and linked erratum. Other retained qualifications concern source-internal patient labels and mixed anatomical

outcomes (W01), endpoint definitions and analysis populations (W03/W08), treatment phases and inconsistent chronology (W07), a single-case observation (W10), and document-version attribution (W12). Remaining not-established judgments concern an isoform-equivalence assertion (W02), preliminary trial wording and report-specific populations (W05), and the binding of a numerical significance threshold to a particular comparison (W09). W11 retains uncertainty about mapping the guideline population to a narrower disease category. These judgments identify interpretation requirements while leaving the benchmark reference sets and performance scores unchanged.

F.4 Validation Scope and Completion Status

Table 36 separates each evaluation target, material unit, completion status, and supported claim. The twelve-family review differs from the eleven-item, single-reader packet for source-judgment development studies, whose results remain in the companion report. The table also distinguishes completed exploratory judgments, the discontinued independent review, and the different completion states of the generated-explanation audits.

Table 36: Evaluation targets and validation status. Completed encoded checks, exploratory reader judgments, and AI-reviewed explanations support different claims. Material counts use the units stated in each row.

Evaluation target	Material and unit	Completion status	Supported claim
Encoded conformance	Frozen predicates and scored trajectories	Complete	Eligibility, provenance, and copied-field compliance under the encoded task.
Publication scope	12 families / witness records; two AI-assisted medical-student readers	Joint judgments complete	Exploratory source-interpretation findings in the selected sample.
Independent source review	30 query–record pairs: 18 random, 12 boundary	Discontinued; incomplete	No completed source judgments or independent validation results.
Generated explanations: fixed exposure	237 outputs in an enriched AI-reviewed sample	Review complete	Exploratory descriptions of generated wording in the reviewed sample.
Generated explanations: live retry	211 items in a prose-audit packet	Unrated	Packet preparation only; no assessed explanation outcomes.
Research workflow	No user study	Not evaluated	No measured completion time, decision quality, or prospective utility.

Supplementary Clarifications and Paired-Outcome Accounting

When Valid Evidence Goes Unused in Oncology Agents

This addendum explains the submitted study and summarizes existing terminal outcomes in a post hoc descriptive analysis. We ran no new model generations. Original scores, primary contrasts, and the status of primary and exploratory analyses are unchanged. The original bibliography and Appendices A to F are included in the same PDF.

S1 Study map and endpoint definitions

The common-prompt study compares valid nonempty inventories under different supplied contexts. A separate inventory replay measures completeness on saved search paths. Case reuse and protocol differences limit comparisons across studies. They cannot be pooled as independent replications or read as successive improvements to a single intervention.

Table 1: Study relationships. Case reuse and protocol differences constrain comparisons across rows. Detailed specifications remain in archival Appendices C to E.

Study	Cases and interaction	Role in the evidence
Earlier feedback pilot	Natural and constructed proposals; original retry protocol	Identifier-only versus diagnostic feedback; preceding exploratory evidence.
Feedback × retrieval crossing	32 co-alteration and 13 sibling-disease positives; four conditions, two repetitions	Supplies the cases reused in both fixed-pool studies below.
Initial fixed exposure	Same 45 positives; ten candidates, two eligible alternatives; no search, one revision; separate negative/pass controls	Co-alteration diagnostic 11/64 versus checklist 10/64; benefit criterion fails.
Common-prompt context	Same 45 positives and candidates; five conditions, two repetitions; clarified instruction and message assembly	Primary fresh 54/64 versus diagnostic repair 43/64; other contrasts exploratory.
Length-matched live retry	Separate 24 co-alteration, 12 sibling-disease, and 12 negative cases; up to two additional searches and resubmissions	Co-alteration diagnostic 30/48 versus checklist 19/48; not an isolated effect of search access.
Inventory and replay	48 cases per arm; recall over 27 positive cases, three repetitions; replay fixes saved search paths	Measures acquired records omitted during finalization.

Valid nonempty inventory rate is the common Boolean endpoint: an answer contains at least one claim, and every claim passes the schema, visible-ID, encoded-eligibility, and copied-field checks. *Valid yield* and *valid recovery* name this endpoint in fresh and post-rejection settings, respectively. Neither implies a complete inventory. **Eligible exposure** records whether at least one eligible candidate became available. **Inventory recall** measures the fraction of the eligible reference set included in the final inventory. The 43/64 valid repair inventories and the 67.2% retrieved recall belong to different experiments and denominators.

S2 Paired outcomes and exploratory heterogeneity

We paired fresh and diagnostic-repair outputs by case and repetition index. Each pair consists of separate generations. It is not a before-and-after measurement of one trajectory, and the shared repetition index does not imply a shared model random seed. This descriptive summary supplements the original analysis, which averages repetitions within cases before variant-cluster resampling.

Table 2: Post hoc accounting of all 90 fresh/diagnostic-repair pairs from the existing common-prompt study. Invalid answers and schema failures remain distinct. Zero cells are retained.

Fresh outcome	Diagnostic repair	Co-alteration	Sibling disease
Valid inventory	Valid inventory	41	4
Valid inventory	Abstention	13	19
Abstention	Abstention	8	2
Invalid answer	Valid inventory	2	0
Schema failure	Abstention	0	1
Total pairs		64	26

The difference of eleven valid co-alteration inventories comes from thirteen pairs favoring fresh and two favoring repair. Eight pairs abstain in both conditions. Thus, not all 21 repair abstentions correspond to a successful fresh output. Fresh also produces two invalid affirmative answers, both blocked by the shared checker. Higher valid yield does not establish a uniform safety advantage. Attempted invalid answers must remain separate from released errors.

Sibling disease shows a larger exploratory difference: fresh 23/26 versus diagnostic repair 4/26, or +73.1 percentage points (reported variant-cluster interval [45.5, 93.3]). Nineteen pairs have a valid fresh inventory and a repair abstention. This stratum includes thirteen cases and eleven variant clusters. Its contrast is exploratory and unadjusted for multiple comparisons. Cases, genes, and candidate content also differ between strata; the result does not isolate a causal effect of disease rejection versus co-alteration rejection.

The primary co-alteration comparison remains +17.2 points [2.8, 33.3], based on 32 cases, 30 variant clusters, two repetitions, and 10,000 paired bootstrap draws. At the case-mean level, seven cases favor fresh, 24 tie, and one favors repair. These case-level counts and the repetition-index pairs describe different units and should not be interchanged.

S3 Generated explanations and submitted actions

The score depends on the structured action and claims, which can disagree with the submitted explanation. In an AI-assisted inspection, the authors selected the following examples from diagnostic-repair abstentions:

- In co-alteration case CQ001, repetition one, the explanation ends by recognizing two candidates as valid, while the action remains **abstain** and the claim list is empty.
- In case B032, repetition two, the explanation recognizes eligible candidates but explicitly cites concern about the preceding rejection when choosing to abstain.
- In case N0443, repetition one, the explanation recognizes encoded eligibility but raises concerns about implicit treatment-population requirements that are not encoded in the inventory predicate.

In these examples, the explanation acknowledges eligibility even though the action omits the candidates. A response can also question the clinical or source-level interpretation of a candidate that meets the encoded inventory rules. Such caution is not necessarily clinically inappropriate.

We selected these examples after inspecting outcomes and have not completed a qualitative census or estimated category frequencies. The examples do not establish an internal causal mechanism. The **reasoning** field contains generated text; it does not expose hidden reasoning. Disagreement between that text and the action does not establish that output-field order caused the disagreement.

A systematic follow-up audit should separate scope misclassification, added requirements, explicit caution after rejection, and final-explanation/action disagreement, retaining an unresolved category. There are 43 diagnostic-repair abstentions across both positive strata (21 co-alteration and 22 sibling disease). An audit could compare these abstentions with their paired fresh outputs, or cover all 180 fresh and diagnostic outputs to include successful submissions. We have not completed that audit or estimated category frequencies.

Existing audits have different targets. Archival Section C reports a 237-item enriched AI-reviewed sample from initial fixed exposure and earlier pilot/crossed explanation audits. The 211 live-retry items prepared for review have not been rated. Section E.5 reports an inventory-task qualitative sample and a post hoc census of ten failed inventories. Those studies do not constitute an audit of the common-prompt 21 co-alteration abstentions.

S4 Prompt sensitivity and identification limits

Diagnostic recovery rises from 11/64 in initial fixed exposure to 43/64 in the common-prompt study on the same co-alteration cases and candidate payloads. Three protocol changes accompany this 50.0-point descriptive difference:

1. The initial proposal appears in both the user payload and assistant call in initial fixed exposure, but only once in the later protocol.
2. The common instruction explicitly permits replacement of a rejected record.
3. It explicitly includes eligible prognostic and N/A-therapy records.

Exact initial and common instructions are disclosed in archival Sections C.4 and D.1. Because the protocol changes were bundled, the comparison cannot attribute the difference to an individual edit or quantify how much of the earlier failure came from underspecification. The recovery rate nevertheless differs substantially between the two formulations. The later fresh versus repair comparison holds the common instruction fixed and addresses a different question.

Fresh receives no erroneous proposal or rejection feedback. Repair receives both as supplied assistant/tool history. This comparison changes the proposal, feedback, message presentation, and input length as a package. Checklist and diagnostic *feedback strings* are length-matched within the specified tolerance; fresh and repair *total contexts* are not length-matched. Reset retains proposal and feedback content as user review material. The diagnostic reset and repair totals are both 43/64, but two case means improve, 29 tie, and one worsens. Equal aggregate yields do not establish equivalence, identical behavior, or validity for naturally occurring errors.

Separating the components requires new controls, such as matched-length neutral context and proposal/feedback presence varied within a fixed presentation. Restarting after removing the rejected ID

is a distinct operational policy: removing one of ten candidates also changes candidate count and input length. A matched removal of an unrelated ineligible candidate would help distinguish rejected-ID removal from generic distractor reduction. We have not run these controls, tested continuations after naturally occurring errors, or replicated the study across model versions in this update.

We froze the common-prompt analysis locally before generating its outputs, after seeing the earlier fixed-exposure results. We reused cases and payloads. Prespecification therefore concerns the subsequent execution and primary contrast, not outcome-independent selection of the research question or an independent case sample. The model is one unversioned alias. Shared publications connect all 45 positives into one component; variant-cluster intervals do not separately account for that dependence. Their coverage for generalization to new publication families is not established. A single connected component cannot supply an informative independent-component bootstrap. Additional repetitions on the same cases do not resolve these limitations.

S5 Deterministic preservation and the interpretation boundary

Let R be the deduplicated records acquired on a saved search path, P the encoded eligibility predicate, and G the full eligible reference set under that predicate. Serialization emits

$$S = \{r \in R : P(r)\} = R \cap G.$$

When identifier handling and field copying conform to the frozen contract, serialized recall therefore equals retrieved recall by construction. The empirical result is the gap between the available eligible pool and the LLM’s final inventory; the identity itself follows from the serialization rule.

On iterative scope-first paths, the LLM retains 847 of 975 available eligible record exposures and omits at least one eligible record in 29/81 positive trajectories. The 128 omitted exposures count unique identifiers within each trajectory. An identifier can recur across trajectories, so this count does not represent 128 distinct corpus records. Case-macro retrieved and final inventory recall are 67.216% and 60.302%, respectively. Their paired difference is 6.914 points, whereas 847/975 is pooled retention and uses a different denominator. Original generation has a 4,096-token response cap with no observed truncations; serialization is uncapped. The replay fixes saved retrieval and changes finalization under these output constraints.

When inventory membership is fully computable, deterministic selection and copying can preserve acquired records without an LLM. This procedure provides a direct baseline for finalization. Search formulation and source-level interpretation are possible LLM roles, but this replay does not establish their necessity, superiority, or clinical benefit. Full-corpus enumeration reaches 100% encoded recall by construction, without establishing publication-level validity.

The checker and reference labels use the same frozen curation. Agreement therefore establishes conformance to the encoded task, but independent publication support still requires source review. W04 illustrates this boundary through observations tied to different genotypes and experimental models. It is one example within an exploratory twelve-family review, not an estimate of corpus error or validation of the repair effect. Two medical-student readers used substantive AI assistance and nonblinded joint adjudication; the separate thirty-pair independent review was discontinued without completed judgments. No workflow study measured review time or decision quality. Preservation and source interpretation remain separate evaluation targets.

S6 Finding material cited in the paper

This guide refers to the 20-page submitted paper. Its condensed appendices and this supplement’s expanded appendices use different subsection and table numbers. All page numbers below are PDF viewer pages in the combined *Extended Methods and Results*. On PDF pages 2 to 32, subtract eight from the printed page number. Tables in addendum sections S1 and S2 have their own numbering.

Sections and tables

Reference in the paper	Expanded supplement	PDF pages
Table 1: protocol roles and budgets	S1 overview; C, Table 14; E.1	33, 13, 25
B, Table 3: samples and limits	B.1, B.5; study settings in C to E	7, 10 to 12
Introduction; A, Table 2: typed scope audit	A.4, Table 4	4 to 5
Section 2; B: shared methods	B.1 to B.5	7 to 12
Section 3; C.1, Table 4: fixed exposure	C.1, Table 15; outcomes in Table 17	13 to 15
Section 4.3; C.2, Table 4: live retry	C.2, Tables 16 and 17	14 to 15
C.3: initial prompts and messages	C.4	19 to 20
Section 4; D.1, Table 5: common prompt and conditions	D.1, Table 22	20 to 21
D.2: design, scoring and statistics	D.2	21 to 22
D.3, Tables 6 and 7: outcomes and contrasts	D.3, Tables 23 and 24	22 to 23
D: detailed resource accounting	D.4, Table 26	24
Section 5.1; E.1: inventory task	E.1 to E.2, Tables 27 to 29	25 to 26
E.2, Table 8: saved-pool serialization	E.4, Tables 31 and 32; replay checks	28 to 29
E.3: controls and omissions	E.3 to E.5	27 to 29
F, Table 9: joint judgments	F.3, Table 35, columns J	31
F, Table 10: W04 observations	F.3, W04 discussion	31
F: source review and its limits	F.1 to F.4, Tables 33 to 36	29 to 32

The initial system instruction, checklist core, and feedback tail appear in C.4 (PDF pages 19 to 20). The exact common system instruction and final command appear in D.1 (PDF page 21). These passages let readers compare the disclosed instructions even though the original prompt JSON files are not included. Related work is in B.6 (PDF page 12), and the bibliography is on PDF page 2.

What the public data reproduce

`data/paired_outcomes.csv` contains 90 fresh/diagnostic-repair pairs (180 outputs): 64 co-alteration pairs and 26 sibling-disease pairs. `source/analyze.py` recomputes the S2 counts. The CSV covers two of the five common-prompt conditions, whose full study comprised 450 outputs. It omits variant-cluster assignments, the other conditions, candidate records, and full responses. It therefore does not support independent response scoring, reconstruction of all five conditions, or reproduction of the reported variant-cluster bootstrap intervals. Original scoring and statistical methods are described in D.2 to D.3.

Files and documents cited in the paper

The paths in the paper and original Appendix B.5 refer to the research workspace used for the experiments. Original Table 13 is a map of that workspace, not this repository. The original experiment directories, code, and execution records are not included here. The passages below explain the methods. Running the original experiments would require the original code and inputs.

Cited file or document	Description available in this PDF
<code>versions/v14-full-paper-revision/README.md</code> ; <code>src/common.py</code> ; <code>src/agents.py</code> <code>analysis_inputs/prompt_material_exact.json</code>	B.1 to B.5, pages 7 to 12: corpus, eligibility, scoring and execution. Original README and code are not included. C.4, pages 19 to 20: disclosed initial instruction and feedback text. Original JSON is not included.
V40 <code>src/study.py</code> ; V38 E1 <code>src/runner.py</code> and <code>src/feedback.py</code>	C.1 to C.2 and C.4, pages 13 to 15 and 19 to 20: repair protocols and message assembly. Original code is not included.
<code>analysis_inputs/v41_public_prompt_contract.json</code>	D.1, pages 20 to 21: exact common system instruction and final command. The full JSON, schema and request hashes are not included as files.
<code>versions/v41-repair-history-control/</code>	D.1 to D.4, pages 20 to 24: study design and aggregate results. Original experiment directory is not included.
Other directories, execution plans, timing files and per-run scores listed in B.5	B.5, pages 10 to 12, and the study appendices. The original files and complete per-run results are not included.
Saved-pool replay code and inputs listed in B.5	E.4, pages 28 to 29: procedure and reported verification. Executable replay and original inputs are not included.
Companion technical report	A, C and F describe related studies and review scope. The separate report and its additional results are not included.
Signed reader submissions and source-bearing review materials	F.1 to F.4, pages 29 to 32: methods and reported judgments. Underlying restricted materials are not included.

Building the public PDFs

`source/build.py` compiles the cover and addendum, combines them with the original bibliography and appendices, and verifies that the original pages have identical text and rendering. It clears identifying PDF metadata and records checks and artifact hashes in `data/verification.json`. The original supplement is retained in `archive/`. The build requires Python 3, PyMuPDF, and `pdflatex`; it does not require the unavailable experiment files, licensed candidate text, full transcripts, or private reader forms. The README lists the commands and links to all public data and source files.